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Review article

Machine learning for fast development of advanced energy materials

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ABSTRACT

With its unique advantages in artificial intelligence, data analysis, interpolation and numerical extrapolation, etc. ML has recently been quickly developed for the discovery of advanced energy materials. In particular, many algorithms have been developed to predict material properties. Herein, we first introduce the ML algorithms used in material science and the structure of each algorithm. Then we examine the algorithms that have been used recently in functional materials, especially in solar cells, batteries, and phase-change materials. Finally, advantages and disadvantages of each algorithm are compared to aid readers in choosing a suitable algorithm for specific applications.

1. Introduction

The majority of material-related data remains unutilized in databases or is only minimally utilized. In this case, developing a fresh approach to research is essential to hasten the pace of material innovation. Currently, material science research is ushering in a new era with the aid of artificial intelligence (AI). After more than 60 years of growth, the primary algorithmic framework and a robust hardware foundation have been established to support AI. This advancement has progressed from a basic perceptron to complicated multilayer neural networks [1]. Some highly advanced AI systems have even successfully defeated human world champions in various fields, such as chess, Go strategy games, and other domains. The field of material science heavily relies on data, and as such, the community places significant emphasis on AI due to its exceptional data-mining capabilities. By utilizing both big data and AI, vast amounts of pre-existing information can be synthesized into untested hypotheses that may guide future research endeavors [2–8]. Meanwhile, this approach is suitable for addressing complex composite spaces or nonlinear processes, which facilitates the resolution of current challenges encountered in material research.

Machine learning (ML) is a critical subfield of AI that has made significant progress in recent years, and it holds the most promising potential for material science research. ML is a method that enables computers to learn how to perform specific tasks by utilizing data already at their disposal [9]. Due to the escalating accumulation of data and advancements in computer technology, ML has recently emerged as a highly sought-after field of study. Furthermore, its applicability spans across a diverse range of contexts [10,11]. When combined with other tools and supplemented with additional theoretical and experimental data, ML techniques can yield more precise and robust results. The generalization capacity and prediction accuracy of ML systems depend on the choice of algorithm [12], which must also take into account the size of the dataset. Datasets used in ML can vary greatly in size, with large datasets often containing complex data that cannot be processed quickly by conventional computers. In the field of material science, classification and regression ML methods are particularly popular [13]. Among these methods, kernel-based algorithms are dominant. The complexity of regression necessitates the use of the kernel method, resulting in the utilization of classification and ridge regression [14] Kernel ridge regression (KRR) is a combination of these techniques and is commonly employed in material science due to its ability to adjust hyperparameters. On the other hand, support vector machines (SVMs) aim to find a hyperplane that best separates the data points in a classification problem [15]. SVMs utilize a maximum margin approach to learning and are often used in conjunction with ridge regression. Nonlinear SVMs can also be applied together with kernel ridge regression [16].

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Nomenclature

AI	artificial intelligence.
ML	ML.
ANN	artificial neural networks.
GA	genetic algorithms.
API	application programming interface.
DFT	density functional theory.
Tc	curie temperature.
SVM	support vector machine.
RVM	relevance vector machine.
RF	random forest.
Sn	Tin.
SHAP	Shapley additive explanations.
XGBoost	extreme gradient boosting.
MAE	mean absolute error.
SVR	support vector regression.
Eg	Bandgap.
PWS	photocatalytic water splitting.
PSCs	perovskite solar cells.
OIHP	organic-inorganic hybrid perovskites.
DSSCs	dye-sensitive solar cells.
Al_2O_3	aluminum oxide.
HTM	hole transfer material.
VBO	valence band offset.
CBO	conduction band offset.
PHM	prognostics and health management.
SEI	solid electrolyte interphase.
ST-PSC	semi-transparent perovskite solar cell.
ITO	indium tin oxide.
IBC	Interdigitated back contact.
NIR	near-infrared.
EVs	electric vehicles.
GBR	gradient boosting regression.
BPANN	backpropagation artificial neural network.
LOOCV	leave-one-out cross-validation.
$CH_3NH_3PbI_3$	methylammonium lead iodide.
GA	genetic algorithm.
NN	neural network.
ASSLB	all solid state lithium-ion batteries.
Li	Lithium.
Na	Sodium.
RUL	remaining useful life.
RMSE	root mean squared error.
NB	naive Bayes.
LSSVM	least squares support vector machines.
XRD	X-ray diffraction.
FTIR	fourier transform infrared.
EA	Ethanolamine.
DE	differential evolution.
SOH	state of health.
RNN	recurrent neural network.
RMS	root mean square.
BMS	battery management system.
SOC	state of charge.
PCM	phase change materials.
BTMS	battery thermal management systems.
LASSO	least absolute shrinkage and selection operator.
ARD	automatic relevance determination.
MEPCMs	microencapsulated phase change materials.
LHTS	latent heat thermal system.
DQN	deep Q-Network.
DSD	dish solar distiller.
CPV/TSD	concentrated photovoltaic-thermal solar dryer.
MC	moisture content (g water/g dry matter).
MR	moisture ratio.
TES	thermal energy storage.
PV/T	photovoltaic/thermal.
TLBO	teaching learning-based optimization.
$E_{out,annual}$	annual output energy.
SOC	State-of-charge.
OSCs	organic solar cells.

With the exponential growth of ML, its application can be broadly categorized into four main domains: predicting material properties, optimizing device structures and fabrication processes, as well as reconstructing measurement data [17]. Several research papers on ML in the field of materials science have been published in recent years to identify current materials' characteristics and functions or to discover new materials with better functions [18]. Machine learning has been systematically employed in the investigation of some typical energy materials/devices such as perovskite solar cells (PSCs) [19–22], secondary batteries, and phase change materials (PCMs). PSCs have emerged as one of the most promising contenders in the photovoltaic community. The utilization of ML in the design process for organic-inorganic hybrid perovskite materials is increasingly prevalent. For instance, Yu et al. [23] created a ML model based on experimental data to describe the compatibility between post-treatment amine molecules and the halide perovskite films and forecast the stability of perovskite films. Moreover, Lu et al. [24] employed the Xgboost method and successfully synthesized six perovskite materials with desired band gaps. In addition, secondary batteries have been widely used in consumer electronics, portable electric devices, and grid-scale energy storage. They can recharge, which makes them ideal for powering high-efficiency technology like mobile phones or electric automobiles. ML will become a viable computational approach for rechargeable battery materials by more quickly and precisely obtaining the composition-structure-property correlations of rechargeable battery materials [25]. PCM is a kind of star materials that can be used in electrification (photovoltaics and wind), high-power-density electronic devices and equipment, energy conversion, electrified transportation, and building air conditioning. An active PCM system is preferable when more accurate system control or higher heat transfer performance is needed [26], it is shown that ML approaches provide excellent potential in predictive model controls on PCMs integrated cooling systems for intelligent charging/discharging of PCMs and innovative ventilation [27].

So in this review, we summarized ML and the algorithms used in ML, then examined the structure of each algorithm and explain the application of ML in the field of PSCs, batteries, and PCMs. Then we examine the advantages and disadvantages of ML in the field of energy material science and provide solutions for better use of ML in the future. Finally, we compare the algorithms used from 2010 to 2023 in energy materials science and compare the results.

2. Machine learning algorithms

ML for materials research is predicated on acquiring sufficient quantities of trustworthy materials data (obtained via experimentation or calculation) that best describe the behavior or qualities of materials or the uses for those materials [28]. This is done in order to generate a model that can be used to use in the process of discovering new materials without having to do the same experiment or calculation again. We are able to uncover previously concealed insights via the use of ML [29]. The techniques of ML may create a model and make predictions once they have learned the rules from a given dataset [30]. Fig. 1(A) provides a synopsis of the many forms of ML now in use. ML methods may be divided into three groups: reinforcement learning, unsupervised

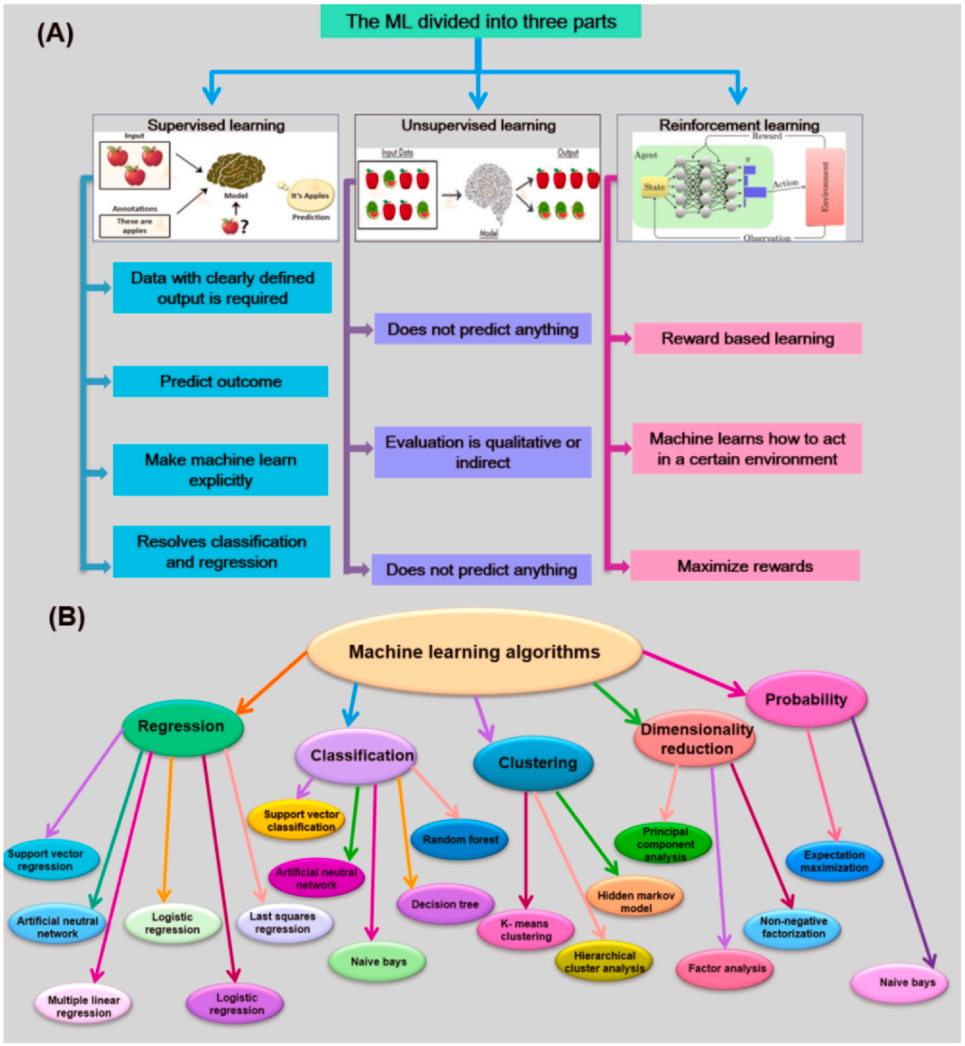


Fig. 1. (A) Different types of ML, (B) Machine learning algorithms.

learning, and supervised learning. Among these, supervised learning is one of the most fundamental approaches of ML, and the process involves an expert analyzing the data [31]. Despite the fact that the data must be correctly labeled for this approach to be successful, supervised learning is a very effective strategy provided that it is used in the appropriate situations [32].

As a subset of the more extensive dataset, the training dataset's primary goal is to provide the algorithm with a high-level grasp of the problem, the answer, and the data points that need to be handled [33]. The properties of the training dataset are extremely similar to those of the final dataset, and the training dataset supplies the algorithm with the labeled parameters that are necessary for solving the issue [34].

After being presented with the training dataset, the program will find connections between the supplied parameters, thereby producing a cause-and-effect relationship between the variables included in the dataset [35]. When the training has been finished, the algorithm will have a broad grasp of how the data functions and the link between the input and the output [36]. After that, it is put into operation for the training dataset. The solution is implemented with the whole dataset, from which the program accumulates knowledge the same way as for the training dataset [37]. This suggests that supervised ML algorithms will continue to develop even after they have been implemented. They will discover new patterns and associations as they train themselves on new data; this will occur because the algorithms will train for new patterns and associations [38].

One of the essential advantages of unsupervised ML is that it can process data that has yet to be labeled [39]. This suggests that humans can do manual work to make the dataset understandable by computers [40]. As a consequence of this, the program will be able to function appropriately with much larger datasets [41]. The labels provide the algorithm the capacity to discern the specific nature of the relationship between any two data points throughout the supervised learning process [42].

In contrast, unsupervised learning does not entirely rely on labels as a foundation for its operation, which might potentially result in the creation of hidden structures [43]. The algorithm has the ability to discern relationships between data points in an abstract fashion, and it does so without requiring any input from human humans [44]. The adaptability of unsupervised learning algorithms results from their ability to unearth patterns and structures that were previously hidden [45]. Learning algorithms that are not supervised can adapt to new information by progressively altering the hidden structures [46]. This is done instead of a preset and fixed issue statement; this gives more options for post-deployment development than supervised learning approaches [47,48].

Reinforcement learning serves as a close model for how individuals learn from the experiences they have and information they gather during their everyday lives [49]. It does this by using an algorithm that, via trial and error, can improve itself and learn from new conditions [50]. Conditioning as a psychological premise is used in the learning

method known as reinforcement learning. It is necessary to place the algorithm in a working environment with an interpreter and a reward system for this method to be effective. The interpreter is provided with the result of the algorithm's output for each iterative process, and it is their responsibility to determine whether or not the results were satisfactory [51,52]. The interpreter will express their gratitude to the algorithm in the form of a reward if the computer program successfully finds the solution to the issue. In the vast majority of cases, the incentive system is inextricably linked to the quality of the outcome. If the output is not desirable, the algorithm will be forced to repeat to locate a more satisfactory solution [53,54].

In frequent applications of reinforcement learning, such as identifying which route on a map takes the shortest distance between two points, the result is not an absolute value. Instead, a score of effectiveness based on a percentage is assigned to show how successful it is. If the value of this percentage is raised, then the amount of compensation that the algorithm receives will also be increased. Hence, the program is trained to provide the best feasible response to get the maximum possible reward [55–59].

Since it significantly affects the accuracy with which the prediction is made, selecting a suitable machine-learning algorithm is critical. There is no one strategy that is superior to all others in every circumstance, and there is a large number of ML software programs [60–63]. The selection of an acceptable algorithm, which is done mostly via a process of trial and error, is one of the most important steps in the development of a highly successful model. It is expected that it may be helpful in choosing the most effective algorithms if the learner has sufficient prior understanding of the algorithms [64,65]. Based on the data type and the question that is being asked, a variety of different learning algorithms may be used (Fig. 1(B)).

2.1. Database

Because the quality of modeling data impacts the accuracy of the ML model, databases play essential roles in using ML for energy material development. According to the material database development history, most extant databases were created in the last two decades. The ML database in energy materials including the Inorganic Crystal Structure Database (ICSD), Cambridge Structural Database (CSD), Open Quantum Material Database (OQMD), and others. Table 1 and Fig. 2 present a selection of the most important material databases. When it comes to the generation of material characteristics, the computing technique plays a crucial role. By utilizing open databases, we can bypass the arduous and time-consuming process of data preparation and production. Users are granted access to an open-source and user-friendly materials API when employing databases generated by high-throughput computations. This is yet another advantage of using such databases.

2.2. Descriptors

The representation of compounds in a dataset plays a crucial role in determining the effectiveness of machine learning (ML) approaches. These representations are commonly referred to as "descriptors" or "features." When conducting ML modeling, it is essential to identify suitable descriptors based on the cost of assessing the target property and the scope of the exploration area. Selecting appropriate descriptors can involve leveraging prior knowledge, expert expertise, and established relationships between the target characteristic and other qualities. However, in many cases, descriptor selection is carried out through a trial-and-error approach. This is because the predictive performance of the model, often measured by prediction error and efficiency, heavily relies on the quality of the target property and the size of the dataset. The majority of candidate descriptors can be categorized into three main groups. Firstly, the limited availability of physical attributes and derivative quantities of compounds in a given library is a crucial factor to consider. Both of these factors hold significant importance. Secondly, the physical characteristics of compounds, as determined by density functional theory (DFT) calculations or derived quantities, contribute to the descriptor set. Lastly, the descriptors encompass the characteristics of individual elements, the composition of compounds, and the quantities of their derivatives. In certain cases, combining descriptors from multiple categories can prove beneficial in achieving improved modeling performance[66].

2.3. Diversity and quality of training data in ML

Machine learning algorithms have demonstrated remarkable performance and have found extensive applications in various real-world scenarios. They possess the ability to adapt to different models and cater to the unique requirements of diverse tasks. The effectiveness of a

Table 1
A collection of popular material data sources.

Database	Description	website
ICSD	The greatest library of fully recognized inorganic crystal structures	https://icsd.products.fiz-karlsruhe.de
CSD	Analyzes intra- and intermolecular interactions inside the lattice	https://www.ccdc.cam.ac.uk/solutions/by-use-case/develop-new-materials/
OQMD	The OQMD is a database of structural and thermodynamic characteristics computed using DFT.	https://oqmd.org
AFLOWLIB	A developing library of material compounds.	https://www.aflowlib.org
Materials project	Open web-based access to computed information on known and expected materials.	https://materialsproject.org
OMDB	The Nordic Institute for theoretical physics – Nordita created and now hosts the organic materials database.	https://omdb.mathub.io
MatWeb	It offers tools for doing searches using data mining and ML methodologies as its foundations. Their online interface offers universal characteristics.	https://www.matweb.com
Altair	Data sheets for thermoplastic and thermoset polymers like nylon, polycarbonate, polyester, polyethylene, and polypropylene are included in MatWeb's searchable database of materials.	https://www.altair.de/total-materia/
CAMEO	Total material includes data on metallic and non-metallic materials, as well as a special collection of advanced characteristics data.	https://cameo.mfa.org/wiki/Category:Materials_database
V2DB	Items used in the creation and preservation of artistic, architectural, archaeological, and anthropological materials.	https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/SNCZF4

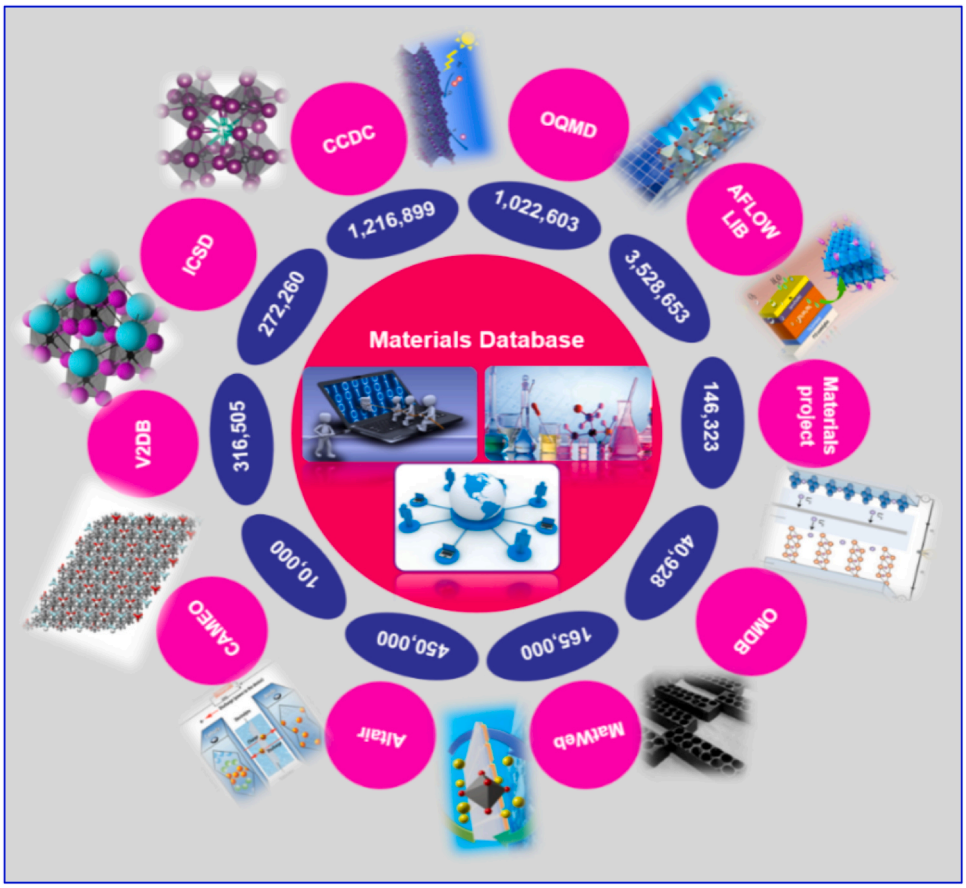


Fig. 2. The last deposited structures in each database.

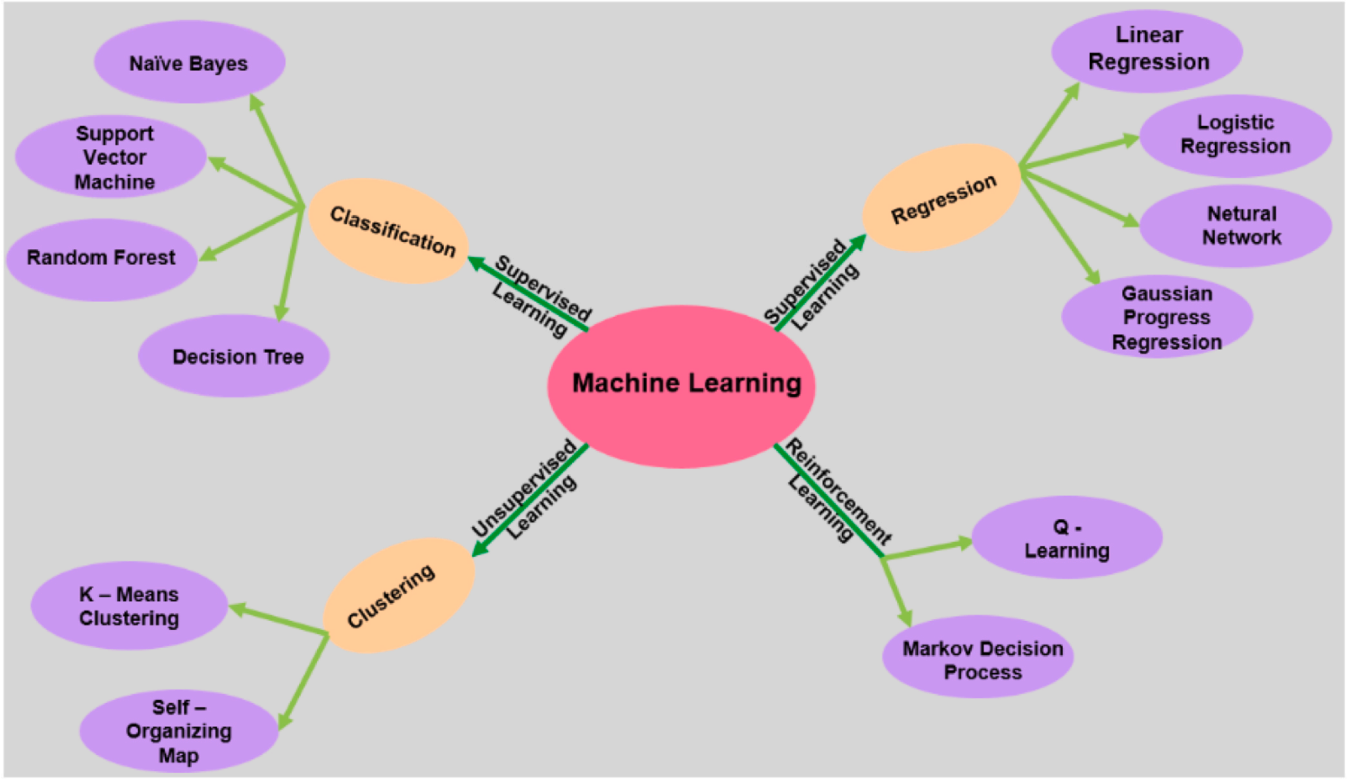


Fig. 3. Common ML algorithms.

machine learning system hinges upon several factors, including the abundance of training data, a well-designed training procedure, and accurate inference. One crucial factor that can significantly impact the performance of the machine learning process is the diversity within the system. Diversity plays a pivotal role in ensuring overall robustness and efficacy. The diversity of training data allows for the provision of more discriminative information to the model, enabling it to capture essential patterns and features. Similarly, diversity in the learned model, encompassing diverse parameters within each model or even diversity among different base models, ensures that each parameter or model can encapsulate unique or complementary information. Furthermore, diversity in inference offers multiple choices, each corresponding to a different parameter or model, enhancing the flexibility and adaptability of the system. Despite the critical role of diversity in machine learning, there is currently a lack of systematic examination and analysis of diversity within machine learning systems [66].

Larger training datasets have the potential to improve accuracy since they provide a greater amount of information to the model. Machine learning models are designed to learn from the data provided during training, and with a larger dataset, the model can detect more intricate patterns and correlations that a smaller sample might have overlooked. Moreover, as the volume of available data increases, the accuracy of the trained model tends to improve. Another advantage of larger training datasets is their ability to mitigate the risk of overfitting. Overfitting occurs when a model becomes overly complex and matches the training data too precisely, resulting in poor generalization to new data. By augmenting the data available for training, the complexity of the model can be restricted, reducing the likelihood of overfitting. However, it is important to note that accuracy does not necessarily increase in direct proportion to dataset size. The presence of noise, outliers, and irrelevant information within a larger dataset of poor data quality can lead to a decrease in model accuracy. Therefore, it is crucial to ensure the accuracy, completeness, and representativeness of the data to avoid such pitfalls [67].

2.4. Types of ML algorithms

Supervised learning algorithms, unsupervised learning algorithms, and reinforcement learning algorithms represent the three primary categories of machine learning (ML) algorithms (Fig. 3). Within supervised learning algorithms, there are two main types of ML models: regression models and classification models [68,69]. These models, which include neural networks and logistic regression, enable the prediction of continuous or categorical outcomes based on labeled training data. Unsupervised learning algorithms, on the other hand, encompass tasks such as clustering and dimensionality reduction [70]. Examples of unsupervised learning algorithms include K-nearest neighbors and principal component analysis, which aim to discover patterns and structures within unlabeled data. Reinforcement learning, a crucial facet of ML, introduces an interactive learning environment where agents learn through trial and error, guided by feedback signals. This type of learning algorithm provides a unique dimension to the field of ML. Prominent examples of reinforcement learning algorithms include Q-learning and the Markov decision process[71]. These algorithms enable agents to make sequential decisions in order to maximize long-term rewards.

2.4.1. Supervised ML algorithms

Supervised learning algorithms represent a common and widely used approach in machine learning (ML), wherein an ML model is trained using labeled historical data (Fig. 4)[72]. In this paradigm, each input variable X is associated with a corresponding target output Y, which can either be a specific data value or a classification label. The key characteristic of supervised learning is that the desired outcome is known for each input variable. Supervised learning ML models can be broadly categorized into regression models and classification models. Regression models are used when the target output is a continuous value, allowing for the prediction of numerical quantities. Classification models, on the other hand, are utilized when the target output belongs to a specific set of classes or categories. These models enable the classification and labeling of data based on learned patterns from the training set. Once the ML model has been trained on labeled data, it can be applied to classify or predict outcomes for unlabeled data. This process involves utilizing the learned patterns and relationships acquired during training to make informed predictions on new, unseen instances. The following sections discuss commonly used supervised learning methods.

2.4.2. Regression

2.4.2.1. Gaussian process regression. Gaussian process regression (GPR) is a highly effective nonlinear approach that enables the analysis of nonlinear systems, even in the absence of prior knowledge about the underlying kernel functions. Unlike traditional regression techniques, GPR provides not only predictions but also estimates of uncertainty associated with those predictions, captured by the variance of the estimate [73]. We can compute the means as the function since we

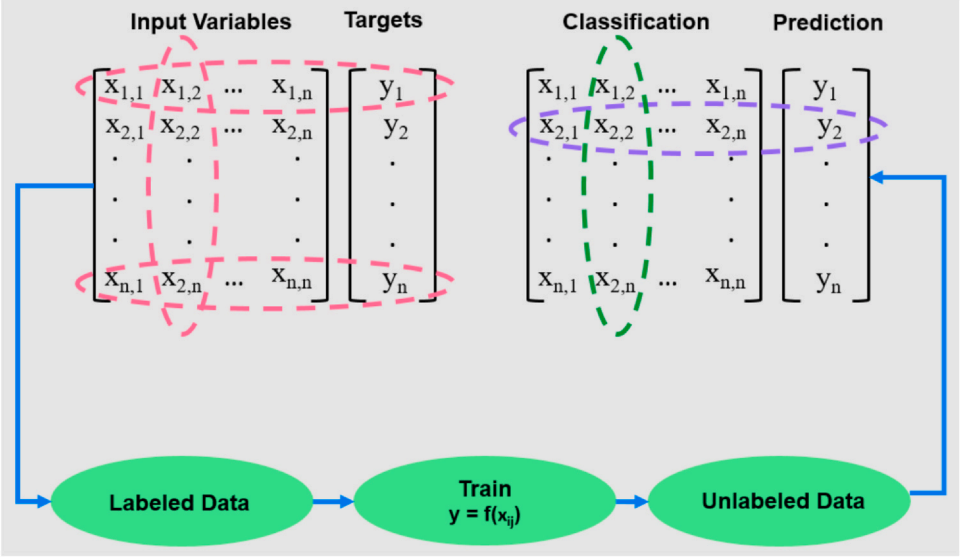


Fig. 4. The process of implementing supervised learning algorithms.

know the probability distribution across all potential functions. The multivariate Gaussian regression function is given as:

$$P(f|X) = N(f|\mu, K) \tag{1}$$

where $X = [x_1, x_2, ..., x_n]$, $f = [f(x_1), ..., f(x_n)]$, $and \mu = [m(x_1), ..., m(x_n)]$are the set of observed data points, m is a function that represents the mean, and K is a function that represents a positive definite kernel. When there is no observation, the mean function is assumed to be $m(X) = 0$ because the data are often standardized to a mean of zero. The Gaussian processes model is distributed over functions, and the form (smoothness) is specified by a constant called K [74]. In general, GPR may be used to forecast the performance of a variety of energy materials. This is particularly true for materials that have complicated structures, interfaces, and compositions, such as lithium-ion batteries and solar cells [75]. In addition, the projected candidates may be eliminated via the screening process in line with the probabilities that were estimated.

2.4.2.2. Linear regression. Linear regression estimates actual values (home cost, number of calls, total sales, etc.) based on constant variables (Fig. 5(A)). Here, we find best line relating the independent and dependent variables [76]. This appropriate linear relationship is known as the regression line and is shown by a linear equation:

$$Y = aX + b \tag{2}$$

where Y is the dependent variable, a is the slope, X is the independent variable, and b is the point where the line passes through the Y axis. Ordinary least squares are the standard linear regression technique [77]. Simple implementation, appropriateness for data sets that can be separated linearly, and regulation to prevent overfitting are all benefits of linear regression. At the same time, the drawbacks of linear regression are its sensitivity to outliers and susceptibility to underfitting. In data-driven materials research, linear regression is often used to assess potential materials and estimate their potential performance [78]. In addition, linear regression methods may be used to quickly and accurately predict outcomes for tiny data sets produced from material tests.

2.4.2.3. Logistic regression. Contrary to linear regression, which produces continuous values, logistic regression is a typical classification process that divides its output into two discrete groups, denoted by 0 or 1. As a result, effective categorization may be achieved. The categorization will be questioned if just linear regression is used, and some of the predicted values of the data points based on the linear regression model may be larger than 1 or less than 0. As shown in Fig. 5(B) putting the data in a logic function predicts the likelihood of an event occurring. Hence, it is called the logistic regression. Because it predicts this possibility, its output values are between 0 and 1 [79]. So, when applied to classification issues, logistic regression may be seen as the promotion of the linear regression model. According to [80], the logistic function is as follows:

$$p = \frac{1}{1 + e^{-z}} \tag{3}$$

where p is the likelihood that landslides will occur, e is an exponential, and z is a linear combination. The following equation illustrates the linear combination "z":

$$z = \beta_0 + \beta_1 x_1 + x_2 + ... + \beta_n x_n \tag{4}$$

where x_i $(i = 1, 2, 3,., n)$ is the number of independent variables, β_i $(i = 1, 2, 3,., n)$ is the slope coefficient, and β_0 is the model's intercept. The expanded version of the logistic regression is as follows.

$$Logit \quad (P) = \frac{1}{1 - e^{\beta_0 + \beta_1 x1 + x_2 + ... + \beta_n x_n}} \tag{5}$$

Logistic regression encompasses several commonly used techniques, including Multi-Logistic Regression (MLR), Ordinal Logistic Regression (OLR), and Binary Logistic Regression (BLR) [81,82]. One advantage of the logistic regression model is its ability to handle both probability and classification tasks. However, it may struggle to capture complex interactions effectively. Logistic regression is frequently employed in high-throughput computational screening to identify high-performance energy materials [82].

Logistic regression exhibits nonlinearity with respect to odds and probability, while maintaining linearity with respect to log odds. Linear regression produces continuous outcomes, whereas logistic regression produces discrete outputs [83,84]. While linear regression aims to find the best-fitting line, logistic regression takes a step further by fitting the line values to a sigmoid curve, resulting in more accurate computations and predictions. One notable advantage of utilizing logistic regression is its ability to quickly identify promising and suitable candidates from new resources, thereby saving time and resources in calculations.

2.4.2.4. ANN. A neural network is a mathematical or computer model that emulates the structure and functioning of biological brain networks [85]. It is capable of solving complex nonlinear modeling problems and making accurate predictions [86] by learning from examples. The most commonly used type of neural network is the Artificial Neural Network (ANN) [87]. Fig. 5(C) illustrates the general algorithmic structure of an ANN, consisting of three layers: the input layer, the hidden layer(s), and the output layer. This layered structure is a fundamental characteristic of ANNs.

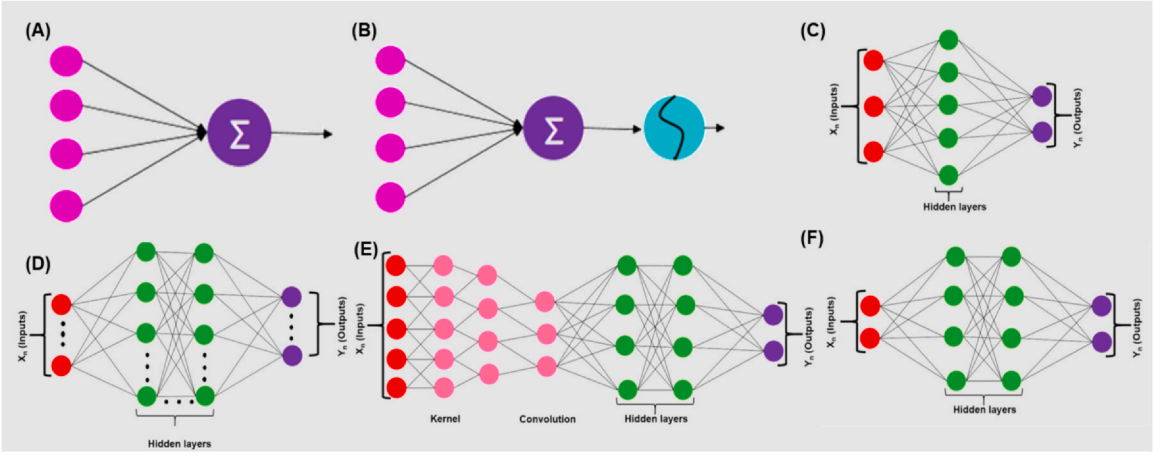


Fig. 5. The architectures of (A) Linear Regression, (B) Logistic Regression, (C) ANN, (D) Deep learning neural network, (E) Convolutional neural network, (F) Recurrent neural network.

The input layer, denoted by the independent variable x, can be modified to include various other variables and integers based on specific feature engineering and domain knowledge in different applications. The input is subjected to a nonlinear transformation performed by the hidden layer, which involves applying weights to the input and passing it through an activation function to produce the output. The internal structure of the hidden layer is modified based on the specific purpose of the neural network. The output layer represents the dependent variable y, which is the target for the neural network's learning process.

ANNs possess several advantages, including fault tolerance, parallel processing capabilities, and the ability to effectively model nonlinear relationships. Deep learning neural networks (Fig. 5(D)), which consist of multiple hidden layers, can represent complex nonlinear systems with even greater accuracy compared to single-layer ANNs. Other types of deep learning algorithms commonly used include Convolutional Neural Networks (Fig. 5(E)), Recurrent Neural Networks (Fig. 5(F)), and Extended Short-Term Memory Networks [88].

Neural networks, due to their ability to capture intricate nonlinear interactions, find applications in solving complex modeling problems. In the field of data-driven materials research, neural network algorithms are employed for predicting material properties and screening potential candidates [89]. Deep learning neural networks, in particular, have the potential to not only predict the interactions between material structure, composition, and performance but also shed light on the underlying principles of chemical reactions. This makes neural networks valuable in the study of materials with complex structures and multiphase reaction surfaces, such as catalytic materials and battery materials. Additionally, neural networks can handle large amounts of data samples, making them a valuable tool when combined with Density Functional Theory (DFT) computations for the discovery of new energy materials.

As shown in Fig. 5**(C)**, the input vector makeup or the same features multiplied by the weights in the inlet vector can be seen. Weights are updated in order; after each example, the weights are updated. The updated amount of the weights are shown below [90].

$$\Delta w_{ij}(n) = \eta \delta_i(n) y_i(n) \tag{6}$$

where η is learning rate, $\delta_i(n)$ is the local gradient and is obtained from Eq. (7).

$$\delta_j(n) = \varphi'_j(v_j(n)) \sum_k \delta_k(n) w_{kj}(n) \tag{7}$$

which j is the input neuron and φ'_j is the derivative of the activation function, which uses the sigmoid function here.

Finally, an error signal is obtained from relationships (7), (8) and (9), and then the error is divided by the number of examples, to produce the mean error.

$$e_j(n) = d_j(n) - y_i(n) \tag{8}$$

$$E(n) = \frac{1}{2} \sum_{j=C} e^2{}_j(n) \tag{9}$$

$$E_{av} = \frac{1}{N} \sum_{n=1}^{N} E(n) \tag{10}$$

where $d_j(n)$ is the desired value, C is a set that contains all the output neurons, and n is the number of examples.

2.4.3. Classification

2.4.3.1. Naïve bayes. One of the supervised learning algorithms using Bayes' theorem as its foundation is the Naïve Bayes (NB) classification technique. Based on the value of the class variables, it is assumed that the conditional independence between each pair of features is naive. The parameters of the Naïve Bayesian classifiers must have a linear connection with the issue's characteristics that must be handled since the Naïve Bayesian classifiers have a high degree of scalability. The use of NB allows for maximum likelihood training to be accomplished by the evaluation of closed-form expressions, which is an approach that is more efficient than the iterative approximation clustering technique. The NB classifier has the potential to be helpful in a variety of real-world settings, even though its assumptions are blatantly oversimplified. The following formula, based on Bayes' theorem, is the one that is most often employed for particular supervised classification issues when the data in question is discrete [91].

$$p(y|x_1,\ldots,x_n) = \frac{p(y)p(x_1,\ldots,x_n|y)}{p(x_1,\ldots,x_n)} \tag{11}$$

As the data set gets more continuous, the assumptions may need to be revised for clustering. It is suggested that the NB algorithm be used as a classification tool to research and develop new energy materials. More crucially, the most recent developments in data-driven materials science show that the Bayes algorithms may be utilized in close-loop optimization to attain material design automation.

2.4.3.2. Support vector regression. Support Vector Regression (SVR) is a regression analysis technique and an extension of Support Vector Machines (SVM) specifically designed for regression problems. It is rooted in statistical theory and offers several advantages over other regression methods. SVM, on which SVR is based, is known for its faster classification process and higher classification accuracy even with a smaller amount of data. However, SVM does not directly provide probability estimates. SVM is not limited to linear data sets and can effectively classify non-linear data by applying dimension modifications using the kernel trick. In the field of materials research, SVM can be applied to classify materials and identify suitable candidates for specific purposes [92]. SVR, a variant of SVM, is extensively used for predicting material properties. It can also be employed in simulating complex dynamic processes, such as the migration properties of lithium ions in lithium batteries. This is just one example of the many applications of

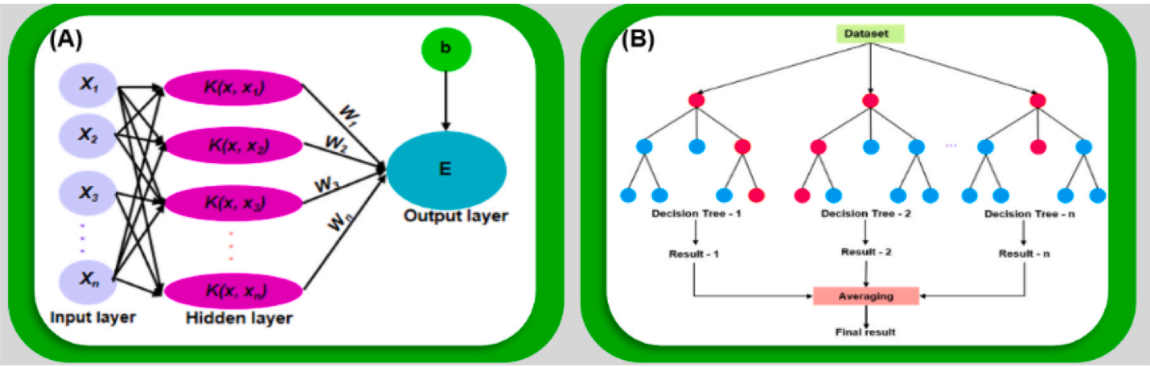


Fig. 6. The architectures of (A) SVR, (B) random forest.

SVR's modeling capability. The structure of SVR, as shown in Fig. 6(A), involves setting a hidden layer that automatically adjusts based on the dataset. This hidden layer establishes the connection between the input layer and the output layer, similar to the design of Artificial Neural Network (ANN) models [93].

2.4.3.3. Random forest and decision tree. Random Forest (RF) is a trademarked ensemble learning algorithm consisting of multiple decision trees, forming a forest-like structure (as shown in Fig. 6(B)). Each decision tree in the forest independently performs classification on new objects based on their characteristics, with each tree "voting" for a particular class. The nonlinear nature of RF gives it an advantage over linear algorithms, making it a popular choice in many applications. However, it is crucial to understand the data and keep in mind that RF is limited in its ability to extrapolate; it can only provide average predictions based on previously observed labels. The final classification decision is made based on the class with the most votes from the forest [94].

Decision Tree (DT) is another learning algorithm widely used in machine learning, primarily for classification problems. DT divides the population into homogeneous subsets based on essential characteristics or independent variables, aiming to create distinct groups as much as possible [95]. The structure of a decision tree, depicted in Fig. 7, typically consists of three components: the root node representing the samples that need to be categorized, internal nodes representing feature characteristics, and leaf nodes representing decision-based classifications. To make predictions using a decision tree, a specific attribute value is used to traverse the tree's internal branches, following the if-then-else rule, until a leaf node is reached, providing the final classification outcome.

Decision trees offer several advantages, including their simplicity, interpretability, and ease of implementation. However, they have drawbacks such as the potential for overfitting due to their simplicity and the instability or complexity of the resulting decision tree structure. Single decision tree models can be easily influenced by noisy data and are prone to overfitting. Decision trees are particularly effective in modeling non-linear relationships. Random Forest leverages the decision tree algorithm to randomly select samples and features [89].

2.4.4. Reinforcement learning and unsupervised learning

Reinforcement learning is a subfield of machine learning (ML) that focuses on learning how to adjust one's behavior based on interactions with the environment to maximize cumulative rewards. One of the most popular reinforcement learning algorithms is Q-learning [96]. Q-learning relies on maintaining a record of the learning process, and this knowledge is utilized by the agent to make decisions that yield the highest returns under given conditions. In the context of the Material Genome Initiative (MGI), unsupervised learning algorithms find significant applications in the categorization of candidate materials into various subsets, facilitating effective classification. Unlike supervised learning, which requires pre-labeled training examples, unsupervised learning techniques can automatically categorize or group input data without prior labeling. Cluster analysis and dimensionality reduction are prominent examples of unsupervised learning applications. Clustering is a common use case for unsupervised algorithms, with the widely used K-means method being a notable example. The K-means clustering technique divides unlabeled data into k clusters, associating each data point with the cluster center that is closest to it in geographical space. Dimensionality reduction, another application of unsupervised learning, can be achieved using methods such as principal component analysis (PCA) and self-organizing mapping (SOM). PCA, for instance, reduces the dimensionality of a dataset, enhancing interpretability while minimizing information loss [97]. SOM, on the other hand, constructs low-dimensional discrete representations of input variables using unsupervised learning. Notably, SOM employs a competitive learning approach rather than an error-correction-based approach and does not include hidden layers [98]. Moreover, supervised learning algorithms and unsupervised learning algorithms can be combined to facilitate the discovery and development of novel materials for energy production and other applications. This integration harnesses the strengths of both approaches to tackle complex material design challenges. Overall, the application of reinforcement learning, supervised learning, and unsupervised learning algorithms presents significant opportunities for advancing material science and enabling breakthroughs in various fields, including energy production.

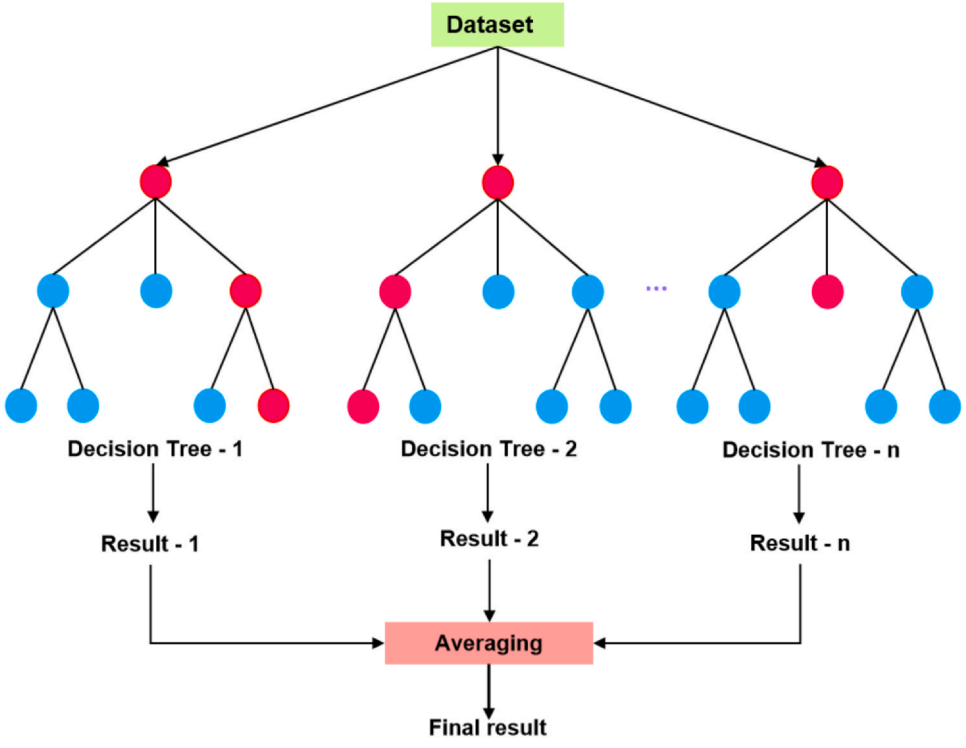


Fig. 7. The Decision tree structure.

3. Application of ML in advanced energy materials

3.1. Perovskite solar cells

Today, increasing energy consumption is inevitable due to the continuing development of human societies [92]. Due to the limitation of the planet's energy resources and the creation of environmental pollution caused by fossil fuel consumption, using solar energy as a clean energy source is an undeniable necessity [99]. Photovoltaic converters (solar cells) are one practical solution for the optimal use of this enormous energy source. So far, much research has been conducted on the design, construction, and optimization of solar cells [100]. In the field of solar cell optimization, ML methods are one category of utilized approaches (Table 2 and Fig. 8) that is both practical and useful. ML is considered a new branch of the field of academic sciences that provides further valuable information by processing existing data [101,102,98]. The fields of application of ML in the design and manufacture of solar cells are divided into three general categories [103,104]: Prediction of solar cell performance, selection of suitable materials, and optimization of manufacturing processes. Organic semiconductor materials are often used in the active layer where light absorption, exciton production, separation, and finally, production of charge carriers is performed. In this structure, at the contact where electrons are collected, a metal with a low work function, such as aluminum, is used. Due to its low work function in this part of the cell, the metal easily collects electrons from the active layer and the transport material [105–107].

Due to the abundance of perovskite formability data, ML has been applied to innovative designs. Valid and trustworthy data should be used to expose material properties as much as feasible.

Hu et al. [109] obtained a PSC with a 23.4% efficiency and excellent long-term stability after conducting four experiments to verify the machine-learning predictions. This PSC exhibited the best stability under these circumstances, as evidenced by the retention of 97.6% of the initial efficiency after 3288 h of aging in the ambient environment. This research demonstrated that using ML to enhance physical semiconductor models is a successful strategy.

Perovskite materials have several vital physical features, including Tc, the temperature at which the second-order phase transition occurs. Before carrying out a significant amount of experimental work, making a rapid and accurate prediction of the Tc of any novel perovskite material is essential. Zhai et al. [110] used SVM, RVM, and RF, to develop prediction models of Tc using the physicochemical parameters. Their findings indicated that each of the three models has excellent accuracy and dependability.

Cai et al. [111] established the relationship between critical parameters and photovoltaic performance in high-profile $MASn_xPb_{1-x}I_3$ perovskite materials by proposing and then realizing a novel machine-learning approach based on a forward-reverse framework. The suggested process created an asymmetrically bowing link between the band gap and Sn composition, which was accurately proven by their tests. This relationship was discovered as a result of the proposed method. The rapid-change zone and the low-band-gap plateau region for small and large Sn composition, respectively, were explained based on a study of structural development and the SHAP library.

Guo et al. [112] proposed a ML framework to investigate the thermodynamic stability and band gap of lead-free halide double perovskites at high speed and precision. This framework required the analysis of the importance of selected features and provided directions for discovering potential lead-free perovskites. Four distinct ML techniques were applied: XGBoost, ridge regression, SVR, and random forest. Regarding thermodynamic stability, XGBoost had the best prediction performance (R^2:0.9935 and MAE:0.0126).

Tao et al. [113], through ML technology, developed structural-property models aimed at higher hydrogen production rate (R_{H2}) and the appropriate Eg to speed up discovering effective perovskite photocatalysts for photocatalytic water splitting (PWS). When comparing the effectiveness of various methods, such as GBR, SVR, BPANN, and RF, the Pearson correlation coefficients (R) of leave-one-out cross validation (LOOCV) were used.

For the $MA_yFA_{1-y}Pb(Br_xI_{1-x})_3$ mixed-ion perovskite, Chen et al. [114] trained a ML-enabled energy model for fast and extensive sampling over the compositional and permutational spaces to map the ion-mixing energies, chemical ordering, and atomic strains. According to correlation analysis, strong lattice distortion in the high-MA/Br concentration regime was shown to be the principal cause of poor device performance. Strong lattice distortion caused high mixing energy, which led to phase segregation and defect development.

Pan et al. [116] developed ML models to describe the photoelectrochemical properties of molecularly engineered halide perovskite materials that were based on $CH_3NH_3PbI_3$ in an aqueous solution. Additionally, the models predicted a complex multidimensional design space for the halide perovskite materials. An experimental photocurrent data set was used to train and validate the ML models. This data set contained 360 data points, each of which reflected a unique combination of dye structures and experimental circumstances.

Jarin et al. [117] utilized various ML models, including RF, SVM, NN, and GA supported neural network (GA-NN) for crystal-structure classification, and SVR and genetic algorithm-supported support vector regression (GA-SVR) models were evaluated to predict the lattice parameters. The accuracy of the prediction model for the crystal structure classification was near 88% on average for GA-NN, and, for the lattice constants regression model, the GA-SVR model yielded 95% on average; this accuracy may be further enhanced by collecting more reliable datasets into the database.

Li et al. [118] constructed a complete feature set of perovskites basing on their chemical makeup and physical structure. That study proposed five descriptors with explicit physical meaning as features to describe the structural characteristics. The Eg was then used as a test

Table 2
The use of ML in the design and discovery of perovskite materials.

Perovskite Structure	Goal Properties	Precision	Algorithms/Database	Ref.
ABX_3/A_2BBX_6	Formability	90%	Materials Project	[108]
3D perovskites	stability	$R > 0.90$	SVR	[109]
$La_{0.66}Sr_{0.3}Ba_{0.04}MnO_3$	Tc	$R_{SVR} = 0.8549$, $R_{RVM} = 0.8549$, $R_{RF} = 0.8352$	SVR, RVM, RF	[110]
$MASn_xPb_{1-x}I_3$	Efficiency, Band gap	RMSE = 0.0386	GBR	[111]
$A_2B^{1+}B^{3+}X_6$	Stability, Band gap	R = 0.9967	RF, XGBoost	[112]
ABO_3	Band gap, Hydrogen production rate	$R_{Eg} = 0.9290$, $R_{H2} = 0.9897$	BPANN, GBR, SVR, RF	[113]
$MA_yFA_{1-y}Pb(Br_xI_{1-x})_3$	Mixing energy	R = 0.9997	ANN	[114]
$A_2B(I)B(III)X_6$	Light absorbing	R = 0.9931	GBDT	[115]
$CH_3NH_3PbI_3$	Photoelectrochemical properties	R = 0.9899	Kriging	[116]
ABO_3	Predict the perovskite lattice parameters	95%	SVR-GA, RF, GA, NN, SVM	[117]
ABO_3	Band gap	R = 0.9350	GBR, SVR, RF, bagging	[118]
$A_xA'_{1-x}PbX_yX'_zX''_{3-y-z}$	Stability, Optical absorption	R = 0.9591	ANN	[119]
ABO_3	High dielectric breakdown	R = 0.9338	XGBoost	[120]

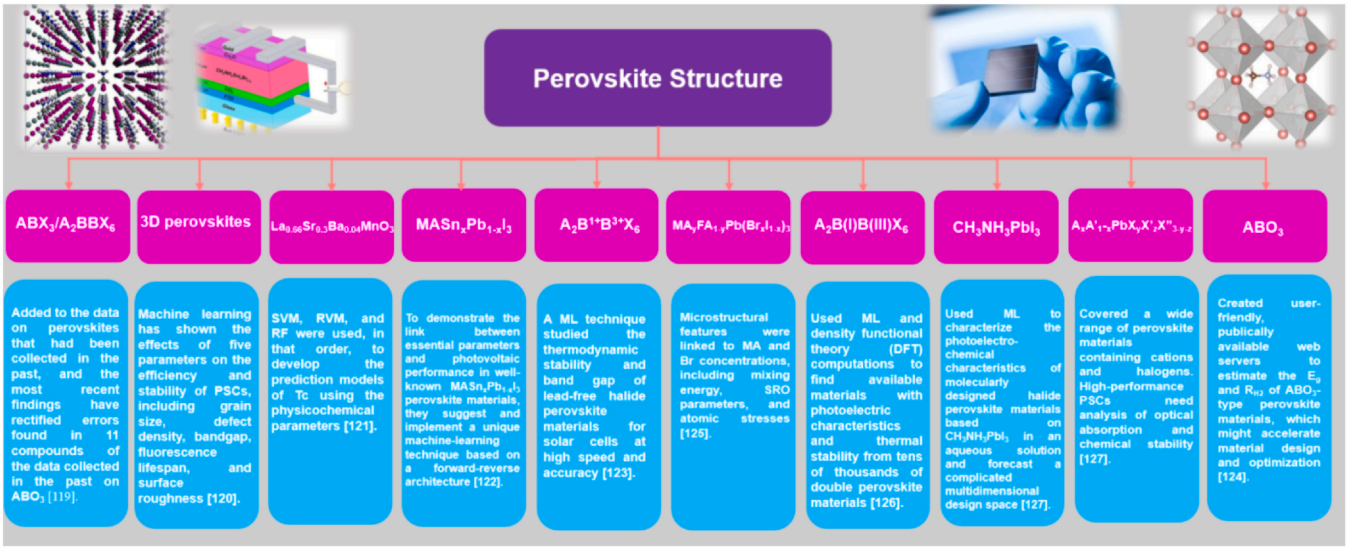


Fig. 8. The accomplishments of ML used in the design and discovery of perovskite materials.

subject for the feature set evaluations conducted by SVR, RF Regression, Bagging Regression (bagging), and GBR.

Filipoiu et al. [119] used DFT calculations assisted by ML techniques to analyze the structurally stable compositions for the maximum absorption possible, while maintaining the best possible level of stability. In this example, the halogen mixture functioned to regulate the opposing tendencies of optical absorption and stability. Similarly, it was discovered that combinations of binary organic cations significantly impact the rate of deterioration. However, they have a more minor, but still observable, impact on the optoelectronic characteristics.

Li et al. [120] predicted the phonon cutoff frequency of perovskite materials as an instrumental variable using properties that can be routinely and readily discovered in materials databases. The performance of this prediction model was exceptional, and it significantly cut down on the amount of computing required. Later, they went through a database of 760 different perovskite oxides and looked for borate perovskite materials with high dielectric breakdown strength. Some examples of these materials include $LaBO_3$, $AlBO_3$, and $LuBO_3$. The findings may assist in the search for potential perovskite materials with high dielectric breakdown strength and potential applicability in dielectric capacitors.

3.2. Batteries

EVs might change the fuel supply, greenhouse gas emissions, maintenance, and driving behavior [121], and it was found that early light-duty EVs reduce air pollution [122]. A nonlinear electrochemical system with inhomogeneous aging is difficult to estimate and forecast [123]. Due to different aging processes [124,125], cell-to-cell variance [126,127], and time-consuming tests [128] especially in field applications—training battery models might be prohibitively expensive. Due to their widespread usage in automotive applications, accurate battery state-of-health (SOH) measurement and prognosis are more vital than ever. However, online diagnostics for real-life EVs are still restricted. All-solid-state lithium-ion batteries (ASSLBs) are thought to be the next generation of energy storage devices [129]. Solid-state electrolytes are an essential component of ASSLBs; thus, they are required to fulfill a number of criteria, including having high ionic conductivity and stable interfacial characteristics [130]. The selection of appropriate materials is the primary source of technical difficulty at the present time in the development of solid-state battery technology [131].

ML can rapidly screen out suitable candidates from large material databases based on the required electrolyte properties, greatly reducing experimental cycles and costs, and effectively solving the problem of difficult material selection. This is made possible by the fact that ML is able to solve the problem of difficult material selection [132]. Major research institutes have created a large number of open-source material databases, such as the Inorganic Crystal Structure Database, Materials Project, and Total Material. These databases include information on energy bands, energy gaps, crystal structures, and fundamental physical property data. The process of selecting materials with strong ionic conductivity may be significantly sped up with the help of ML models, which can discover materials 1000 times more quickly [75,133].

3.2.1. Prediction of the material characteristics of batteries

There are potentially thousands of compounds that can be synthesized for use as battery materials. These compounds can be created by either adding metal ions (Li, Na, etc.) or changing the molar fraction of the elements that make up the compound. Many of these compounds have not yet been synthesized. Because it is already challenging to describe the materials, researchers in the field of batteries need to incorporate the results of their prior experiments in order to more correctly forecast the features of new materials [50,134].

The thermodynamic stability of arbitrary chemicals, the binding energies of multiple compounds, the melting temperatures of simple component solids, the coordination energies of alkali group metals in battery electrolytes, and the ionic conductivity of electrolyte materials can all be predicted by ML models. ML models can also predict the electrode voltage of lithium-ion batteries [135,136].

3.2.2. Conventional ML algorithms for battery applications

Used Naive Bayes to estimate Li-ion battery remaining useful life (RUL) across discharging cycles and operational circumstances. Their RUL prediction methodology used classification methods and is simple and efficient. Different batteries and operating cases provided stable and reliable RUL forecasts. Performance assessment used public Li-ion battery data, RUL at four-time intervals, and RMSEs. The NB outperformed the SVM capacity variations throughout 168 lithium-ion battery charging and discharging cycles. The experiment used normalized battery data. The experiment predicted the remaining 143 cycles using the first 25 cycles of data; RUL prediction references are for the last 143 cycles [137].

The RUL prediction of a battery is an efficient method that can be used to increase the battery's dependability and service life. Based on a solid tracking particle filter, logical regression is a new assessment technique of battery states proposed in this study for RUL prediction

using batteries. The fundamental concept behind this approach is to provide an approximation of a process that is neither linear nor Gaussian. This will be accomplished by combining logistic regression with the least square support vector machine. The two primary contributions are: first, they combined logistic regression with the least square support vector machine for RUL estimation; second, they introduced logistic regression with particle update using a strong tracking particle filter. Both of these contributions are described further below [138]. Using XRD, FTIR, and EA, measured samples and obtained 9 material property values as predictor variables for each piece. Rate and cycle performance tests were performed on 11 $LiFePO_4$ electrodes to compute low rate capacity, rate performance, and cycle life performance ratings. They built multivariate linear regression models based on 9 material variables to predict electrochemical characteristics using least squares estimates of regression coefficients. From the coefficients of determination, R^2, of related regression models, they verified that their regression models are practical and valuable for prediction [139]. Customers who utilize battery-powered items place a high premium on an accurate prediction of the remaining functional life of the battery. This research aims to construct a model for estimating the remaining usable life of a battery pack via k-nearest neighbor regression. The model will include data from each cell in the battery pack. A method known as differential evolution is used to maximize the parameters found in the estimating model. The remaining useful life of a cell is calculated using this method by taking a weighted average of the useful lives of many cells that are physically close to it and have a degradation pattern comparable to that of the cell whose remaining useful life has to be assessed. The newly developed approach estimates the remaining usable life with a standard deviation of 9 cycles, whereas the most accurate calculation has a standard deviation of just 2 cycles. Within a span of ten milliseconds, each of these estimates is carried out. Improving the accuracy of the estimate requires testing more cells and using the ones that are closest together. Compared to particle filter and support vector regression, the newly proposed approach lowers the estimated average error by 83.14% and 89.79%, respectively. Consequently, the estimated findings and comparison confirm the approach's usefulness for estimating the remaining usable life of lithium-ion batteries [140].

An innovative method for evaluating one's current health state was suggested. In the model, the only variables that were considered to be independent were the physical characteristics that were associated with the functioning of the battery. The state of health was considered to be an unknown variable. The estimate of SOH has improved thanks to the data model based on the live process. This model ensures a greater level of consistency with real applications. This study first selected numerous features from the raw data, such as current, voltage, and time during the charging and discharging of the battery operation. Next, it obtained the weights of each feature through the initial training of gradient boosting decision tree (GBDT), and finally, it screened them by combining the correlation coefficients and variance inflation factors (VIFs). In the end, the iteration approach was employed to obtain a combination of characteristics that had outstanding performance. The NASA dataset shows that the trained model can reliably forecast the residual capacity of batteries whose exact characteristics were unknown. This project's most significant contribution is the automated, unguided selection of beneficial characteristics rather than the manual, guided selection that has traditionally been done. At the same time, by screening the multidimensional properties, the approach may help identify the key factors and their processes that impact the life of a battery [141]. Because of the high computational cost, the coupled electrochemical-thermal model cannot be used in the simulation-based battery design. In this research, they proved that neural networks are beneficial for the design of batteries, therefore lowering the amount of work that has to be done via calculation by many orders of magnitude. The information obtained from running finite element simulations was used in building two neural networks, as well as their training and validation. The very first neural network, known as the classifier, was used to determine whether a given collection of input data is really possible. Calculations of specific energy and power were made with the assistance of the second neural network, which was called the calculator. The results of the comparison between the neural networks and the finite element simulations revealed that both neural networks attained satisfactory accuracy levels [142].

When the random forest method is used to forecast the remaining life of a battery, the outcome of the prediction is unpredictable, and it is challenging to ensure that the model is accurate. In light of the issues mentioned above, the authors of this work suggest using evolutionary algorithms to improve the random forest prediction model. The depth and number of decision trees in the random forest are optimized to guarantee the accuracy of the prediction. The result of this process is utilized for life prediction, which uses the ideal combination of decision tree depth and number. Conducting simulation trials and evaluating the

Table 3
Typical ML algorithms employed within battery applications.

Algorithm	Performance	Application	Ref.
Naïve Bayes	Efficiency in classification that is stable, strong performance even with tiny data sets.	Estimating the lifetime of a lithium-ion battery	Selina [137]
Logistic regression	Easy implementation, little computing work, exceptional speed, and minimal use of storage resources	Prediction of the total usable life left in a battery cell	Liu[138]
Linear regression	Simple to use, straightforward to calculate	Analyze the relationship between the material property parameters and the electrochemical performance of lithium ion batteries and make a prediction.	Cheng [139]
K-nearest neighbor	Using nonlinear classification with great accuracy and no data assumptions; seasoned theory	Create an estimate of the product's remaining useful life	Zhou[140]
Decision trees	Being capable of handling uncorrelated characteristics and better adapted to handling uncorrelated features.	Evaluation and shortening of the duration of test batteries that are used for categorization	Zhang [141]
Artificial neural networks	Can handle nonlinear feature interactions, solve high-dimensional problems, and generalize more effectively.	Factors in the design dependent on the available energy capacity	Wu[142]
Random forests	Both regression and classification are possible to carry out.	Battery life expectancy calculations	Cailian [143]
Convolutional neural networks	Sharing of parameters and using sparse connections both lead to a substantial decrease in the total number of training parameters.	Prediction of the capacity decline curve for a battery using early life data	Saxena [144]
Recurrent neural networks	The content of sequences may be modeled using deep models in the time dimension.	Prediction of the deterioration of the battery in light of unknown future situations	Lu[145]
SVM	Problems of a high dimension may be solved; interactions between nonlinear characteristics can be managed; generalization can be improved.	Determine the amount of charge remaining in the battery.	Anton [146]

model's prediction performance using data on lithium-ion batteries that NASA provided, and then comparing the findings of those evaluations with the prediction results of the random forest prediction model and the lasso prediction model [143]. Utilization an open-access experimental graphite/$LiFePO_4$ battery dataset (178 cells) and demonstrated the effectiveness of a convolutional neural network architecture in predicting the entire capacity fade degradation curve using just the first 100 cycles of data. The developed modeling approach utilizes a bilinear equation for the capacity fade of batteries, which can describe both linear (without rollover) and bilinear (with rollover) behavior, and hence can be easily applied to other battery chemistries not covered in the dataset [144]. Using the extensively researched RUL prediction to predict charge and discharge rate trajectories under fixed and random future operating circumstances; this was accomplished by creating a universal deep-learning framework centered on a RNN that took inputs from a current plan and a few early capacity-voltage data points. As a case study, they cycled 77 commercial batteries under fixed and random operating circumstances [145]. Implemented an SVM model on a BMS system based on a microcontroller that inexpensively performed the function of SOC prediction. It may deliver precision equivalent to that of more complex approaches while reducing the amount of computing required.

The SVM model was evaluated to make correct predictions about the cell voltage under dynamic settings, and it did this by achieving a coefficient of determination of 0.97. The maximum error in voltage was held to less than 4%, and the root mean square error (RMSE) for the whole test was held at 0.4% [146].

According to reported data, ~ 33% of published papers regarding ML from the years 2018–2023 utilized the SVM algorithm because SVM is an innovative learning machine that effectively maps nonlinear functions based on statistical learning theory. Because a battery is a nonlinear system, it is challenging to determine the connection between the load voltage and the current at varying temperatures and safe charge levels. SVM is used to describe the nonlinear dynamics of the battery. Table 3 illustrates the ML algorithms that are typically used in ML and their applications in the battery field.

3.3. Phase-change materials

Phase-change materials (PCMs) with high latent-heat capacity can be integrated with building energy systems to save energy without compromising the occupants' needs (thermal comfort, indoor air quality, illuminance, etc.). They can be used with various energy sources (natural cooling [147], water-based thermal energy [148], nocturnal sky radiative cooling [149], solar-thermal energy [150]) (solar-to-electrical, solar-to-thermal [151], solar-to-hybrid thermal and electrical [152], and air-cooled solar PV conversions [153]).

PCMs have been increasingly used in various applications, including electronics cooling, battery thermal-management systems (BTMS), building components, thermal energy storage in solar power plants, heat exchangers, and waste-heat recovery systems [154–156]. Table 4 summarizes the application of ML in the design and discovery of PCMs in recent years.

Table 4
Application of ML in the design and discovery of PCMs in recent years.

Algorithm	Performance	Application	Ref.
RNN	The enthalpy porosity formulation simulates PCM phase transitions to train the network. For PCM liquid fraction prediction, the RNN is evaluated using heat fluxes outside the training data.	To serve as a diagnostic tool for thermal management systems, a RNN has been built to predict the liquid fraction time series of PCM using surface temperature time series as input.	Anooj[157]
ANN, GA	240 numerical solutions trained the neural network to trend input-output correlation. The findings indicated that thermal insulation installation position strongly affects PCM installation placement.	A PCM layer was installed in the building, and investigations into energy use were carried out in the heating and cooling sections.	Abbasian-Naghneh [158]
KNN, ARD, LASSO	The thermophysical characteristics of three-nanoparticle PCM were predicted using ML approaches. Nano-enhanced PCM included MWCNT, GNP, and NG.	This study helps solar energy storage researchers choose the optimum approach for this innovative PCM.	Gao[159]
ANN	Three Reynolds values between 100 and 300 and volume fractions from 0% to 10% in different geometries are investigated. Thermo-hydrodynamic parameters show the outcomes.	This work investigates plate battery cooling with chaotic flow and microencapsulated phase transition materials.	Fini[160]
ANN	CFT's potential was maximized using ML. Temperature transients from PCM melting tests are used to test the capacity of this deep learning approach (multi-layer perceptron model or "MLP") to estimate the time needed to achieve the PCM melt-fraction.	This research accurately predicts melt-fraction time.	Ren[161]
TLBO	In terms of technological feasibility, a hybrid renewable system with thermal and electric exergy forms, latent and sensible heat storage, and active cooling solutions was analyzed.	To resolve the paradox and maximize exergy using an innovative optimization procedure.	Tang[162]
DQN	DQN reinforcement learning was utilized to handle the complexity of optimal control of a binary variable system.	To discover the best energy management method for a system with a solar collector, PCM heat source, and backup heater.	Ebrahimpour[163]
ANN	A convex dish absorber was suggested instead of a flat absorber liner. A circular stepped surface replaced the flat absorber surface.	ML algorithms projected the water production of three existing solar distillers.	Bamasag[164]
ANN, SVM	Using nano-enhanced PCM, a full photovoltaic-thermal solar dryer (CPV/TSD) is experimentally studied.	Two ML algorithms predict the drying parameters and compare the prediction success with four evaluation metrics.	Karaağaç[165]
ANN	Heat flux changes from varied heat source operation were the main problem. A practically constant-temperature molten salt phase transition material was employed for this.	ANN-controlled PCM thermal energy storage units moderate heat flux variations in this work.	Tietze[166]
ANN	Supervised ML and heuristic optimization methods created a general optimization technique. Multivariable optimizations were performed in five Chinese climates for wider use.	Solar-to-electricity conversion, active water-based and air-based cooling, and distributed storages were some of the advanced energy conversions and multi-diversified energy forms that were demonstrated as part of an integrated hybrid system that involved phase change material and advanced energy conversions.	Zhou[18]

Diagnosed thermal-management systems using a RNN to predict PCM liquid fraction time series using surface-temperature time series. The RNN model was trained using continuous, pulsed, randomly rising, and decreasing heat fluxes to cover most latent heat thermal system (LHTS) heat-flux boundary conditions. The enthalpy porosity formulation simulated PCM phase transitions to train the network. For PCM liquid fraction prediction, the RNN was evaluated using heat fluxes outside the training data [157]. Using ML to analyze PCM to improve the thermal resistance of a building wall and found two crucial inputs: the interior temperature of 20–27 °C, and PCM installation position. This is significant since the material may not change phase if not appropriately placed, and the substance's favorable benefits may be negated by not changing its phase [158]. ML approaches were devised and evaluated for estimating, including KNN, ARD, and LASSO. These early algorithms created formulae and models for the problem. Thus, the R-squared values of each constructed ML model were reported, and some recommendations were made to improve nano-enhanced PCM efficiency in thermal systems. This study helps thermal solar-energy storage researchers choose the optimum approach for this innovative PCM [159]. Created and proved the effectiveness of attaining real-time prediction of the actual time left to achieve a preset melt fraction (selected by the user) for a candidate PCM. This was accomplished for a candidate PCM. In this research, a technique that is based on ML and makes use of ANN was investigated with the purpose of assessing the usefulness and dependability of said approach for various kinds of training data [161]. A PCM-integrated renewable system designed to study the active cooling options to increase energy efficiency. The conflict between active-cooling facility power use and renewable production was examined and resolved. A general ML-based optimization technique was used to overcome energy contradictions and maximize the energy of the hybrid renewable system with various active cooling solutions and enhanced solar-to-electricity and solar-to-thermal conversions [162]. A concentrated photovoltaic-thermal solar dryer CPV/TSD developed that incorporated a nanoparticle-doped PCM of paraffin and 5% Al_2O_3 nanoparticles. ANN and SVM algorithms predicted moisture content (MC) and moisture ratio (MR) parameters well. ANN performed slightly better than SVM when all evaluation metrics were considered [162].

A hybrid PV/T-PCM system with a maximum equivalent overall output energy of about 218 kWh that was more technically feasible and cost-effective in Kunming and Haikou compared to Shanghai and Hong Kong in terms of the optimal design and reliable operation of the proposed hybrid system in different cities was proposed. The poor on-site electricity generation in Shanghai was the cause of the low equivalent overall output energy. In contrast, the intense heat transfer in the PV/T-PCM system in Hong Kong was the cause of the low equal overall output energy [18].

4. Advantages, and disadvantages of each ML algorithms in materials science

A model for examining materials may be built with enough data in a suitable format. The modeling process includes selecting proper algorithms, training using training data, and producing correct predictions. In the following discussion, four types of ML are considered: supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning.

Supervised learning is sometimes known as "learning with a teacher" since the training data's associated outputs have been labeled. In contrast, with unsupervised learning, the corresponding outcomes of the training data are unlabeled. In semi-supervised learning, part of the training data is labeled while the remainder is unlabeled; the unlabeled data often outnumbers the labeled data. Instead of instructing the model on creating actual actions, reinforcement signals offered by the environment are utilized to assess the quality of the produced activities and enhance the techniques for adapting to the environment. Table 5 and Fig. 9 summarized the accuracy, errors, and advantages of each ML algorithm in recently published papers, and Fig. 10 illustrates the percentage usage of each algorithm in recently published papers. There are algorithms available for implementing the four kinds of ML techniques outlined above, which may be classified into two groups: deep learning and shallow learning. Table 6 compared each algorithms advantages, disadvanges and applications in advanced energy materials.

Ibrahim et al. [167] proposed that solar radiation in Perlis, located in Northern Malaysia, may be estimated using the linear regression model. The average air temperature and the amount of solar radiation have a linear connection. When no outliers exist, the linear correlation coefficient between air temperature and solar radiation is 0.7780. It drops to 0.7473 when outliers are included. Based on these findings, the correlation coefficient and the value of R^2 are increased when the outliers are taken into account.

Jagadeesh et al. [168] utilized the accuracy attained for the quantity of data stream investigated and the likelihood prediction to anticipate the production from a solar plant. They found that proper variables must be chosen for effective forecasting of solar plant production and fed into the logistic regression method to achieve efficiency in training the data and predicting plant output. Azza et al. [169] described a ML algorithm's prediction of future solar cell photocurrent. They evaluated and studied several prediction algorithms before deciding on one. Because of their adaptive and probabilistic model selection process, precision, MSE, and MAE were chosen as metrics. As a research approach, this work uses ML algorithms to create models for forecasting solar cell photocurrent. They describe the ML algorithm's prediction of future solar cell photocurrent. They evaluated and studied several prediction approaches before settling on decision trees. Antón et al. [170] estimated the SOC of a high-capacity lithium iron phosphate ($LiFePO_4$) battery cell from an experimental data set obtained in the University of Oviedo Battery Laboratory (UOB Lab) using an SVM approach. The SOC of a battery cannot be measured directly and must be estimated from measurable battery parameters such as current, voltage or temperature. An accurate predictive model able to forecast the SOC in the short term was obtained. The agreement of the SVM model with the experimental data set confirmed its good performance. Harippriya et al. [172] presented a battery management system for predicting the remaining battery charge of an electric vehicle. Several machine-learning and

Table 5
The accuracy and errors of each ML algorithm in material science.

Algorithm	Application	Error	Accuracy
Linear regression[167]	Calculate the amount of solar radiation	–	0.6054
Logistic regression[168]	Solar irradiance	–	90%
Decision tree[169]	Forecast of photovoltaic cell Iph	1.231507e-05	0.617507
SVM[170]	High-capacity battery cell SOC prediction	–	0.97
SVR[171]	Energy gap forecast	0.07	0.94
Naïve Bayes[172]	Forecast battery capacity	–	88%
Random forest[173]	Forecast PCE of organic solar cells	0.047	0.80
GBM[174]	Find a set of characteristics used to categorize 5329 ABO_3 compounds	–	80.3%
ANN[175]	Predict the 2-terminal tandem cell's $E_{out,annual}$ with varied perovskite thicknesses bandgaps	–	0.9992
RNN[176]	Forecast solar radiation	–	0.983

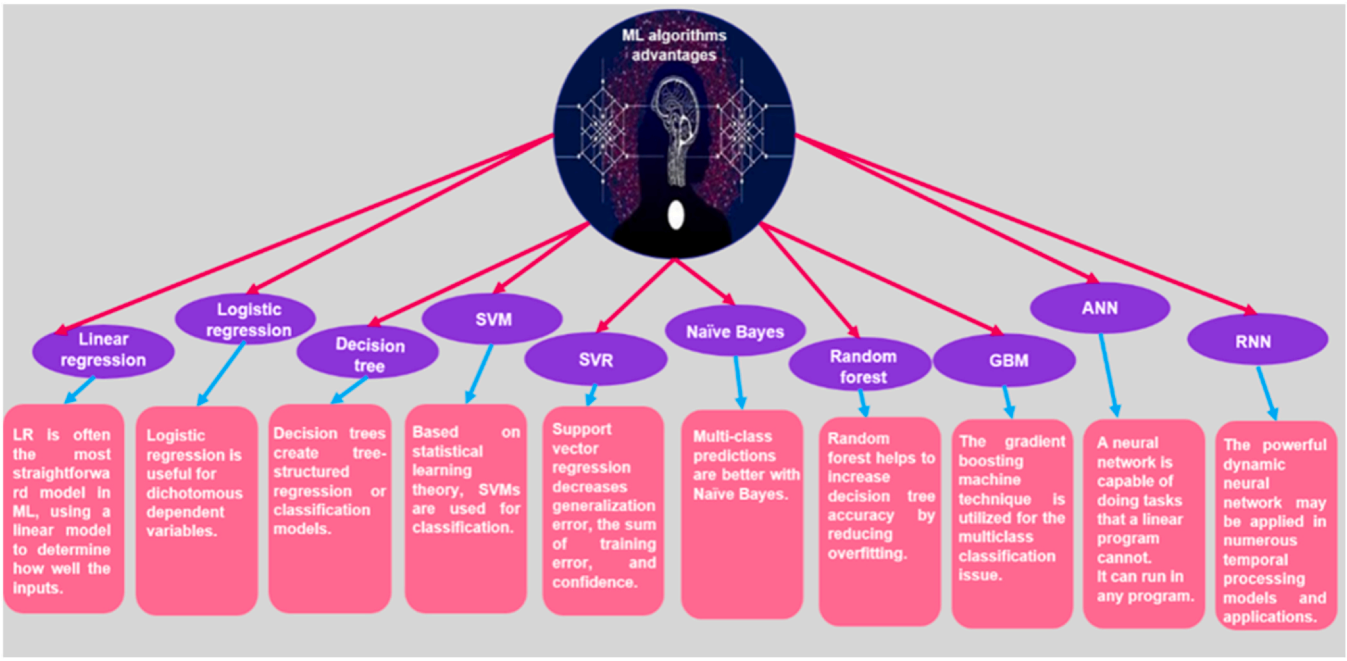


Fig. 9. The advantages of each ML algorithm in material science.

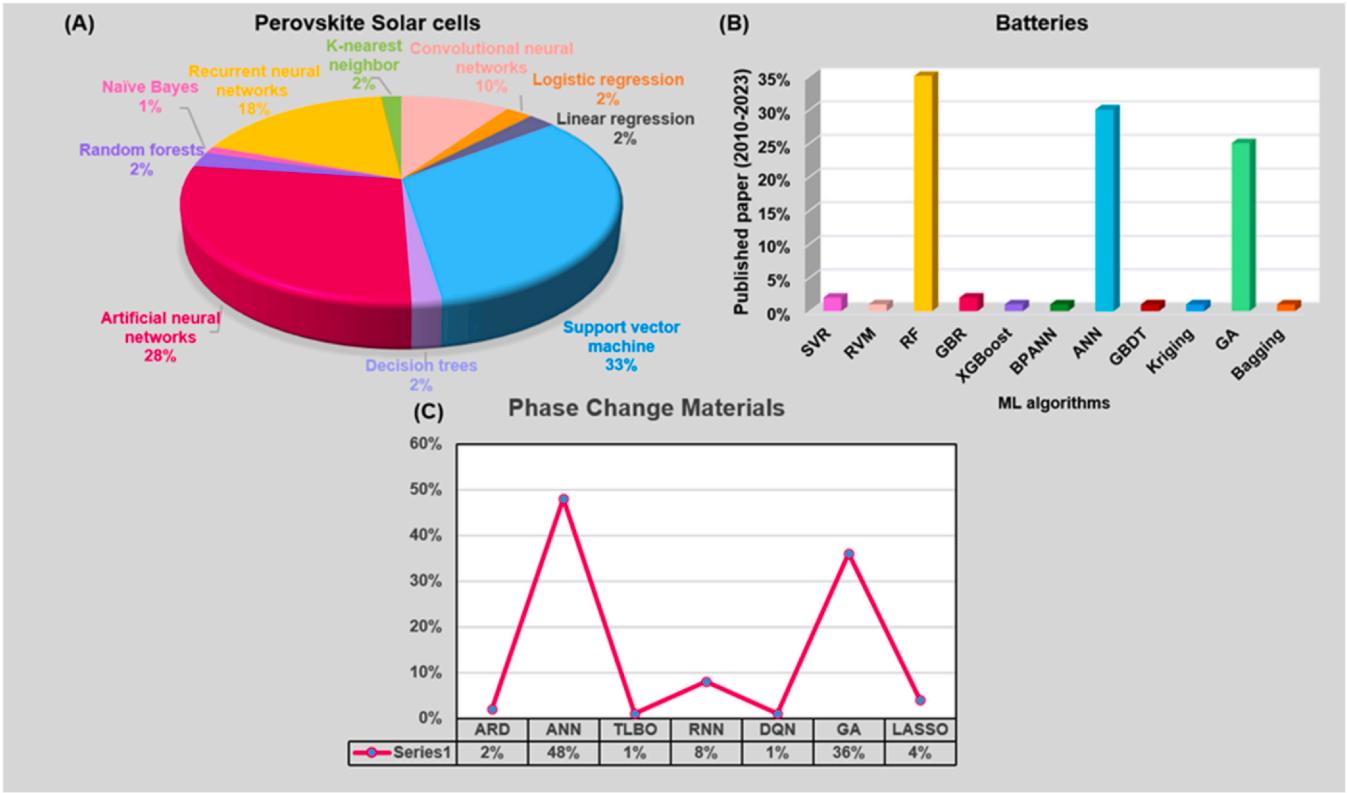


Fig. 10. The percentage of recent studies that used each algorithm for (A) perovskite materials, (B) batteries, and (C) PCMs.

deep-learning techniques were used to anticipate the aging of the lithium-ion (Li-Ion) battery included in the electric car. The parameters such as voltage, current, and temperature were derived via ML methods such as Naïve Bayes. Lee et al. [173] utilized a homogeneous ensemble model dubbed the RF method for the prediction of PCE for non-fullerene organic solar cells (OSCs). The database of > 100 donor/non-fullerene acceptor bulk heterojunction photoactive layers compares two machine-learning techniques (RF and GB) for PCE prediction of non-fullerene OSCs. For quantitatively predicting PCE, their RF model with five readily available descriptors achieved coefficients of determination (R^2) of 0.85 and 0.80 for the training and testing sets, respectively.

Behara et al. [174] used a Light GBM model to classify 675 compounds out of 5329 ABO_3 perovskites. As a result, the model may be utilized as a preliminary, quick, and low-cost approach to organizing perovskites into their various crystal systems (i.e., cubic, tetragonal, orthorhombic, and rhombohedral). The forecast accuracy may reach 80.3%. In addition, the feature importance graph and Shapley value analysis were utilized to determine the link between the characteristics

Table 6
Compression of ML algorithms in energy storage materials.

Algorithm	Application	Advantages	Disavdantages
GPR[177]	Prediction of properties	Organizational simplification, fewer variables, and an understandable probability equation	A significant amount of computations
NN[178]	Prediction of properties and structures, and also material testing	Abilities for efficient nonlinear simulation	It is difficult to describe the model since it is not apparent
RF[179]	The evaluation of materials and the forecasting of their qualities	It can process high-dimensional data and quickly provides the outcome of feature analysis	Data that is sensitive to noise and has a significant computational price
SVM[180]	Material selection and estimation of properties	More precision in classification and a quick classification procedure	No direct probability estimations available
Linear regression[181]	Materials selection and property prediction	Simple application and comprehension	Sensitivity to errors and susceptibility to underfitting
Logistic regression[182]	Prediction of properties and material testing	Valuable for predictive and classification models	Capturing intricate interactions is challenging
Naïve Bayes[183]	Successful optimization	Steady classification effectiveness and provides quickly understandable results	Bayesian probabilities must be calculated, and the expression of the incoming data must be considered.
Decision tree[184]	Material selection and property estimation	A straightforward algorithm framework that is easy to understand and use	Simple overfitting and a tree structure that is capable of instability
Principal component analysis[185]	Size reductions	It simplifies the data set's utilization, decreases computation costs, and contains no parameter limitations	Information loss is minimal, although information overlaps can't be removed entirely

employed and the projected crystal structures. Given the provided Shapley values, more work might be done in which features are designed to pursue a desired crystal structure. As a result, their study is directly applicable to the rational design of perovskites.

Nguyen et al. [175] optimized a tandem structure design by presenting a deep-learning strategy to forecast the yearly output energy captured by the solar cells of the 2-terminal perovskite/silicon tandem; this allowed them to maximize the efficiency of the harvested energy. The simulation results of the tandem cell with the various perovskite layers' bandgap and thickness under real-world conditions including the solar spectrum, incident spectral angle, and solar module temperature in a particular month of the year for a specific direction made up the data set used for training and validating an ANN; these results were obtained by running the Atlas simulation. Consequently, they demonstrated that the solar spectral shape considerably affects the performance of the ANN. The presented ANN model had an MSE of 1.26 and a correlation value of 0.99979.

Pang et al. [176] reported that RNN-based deep learning solar prediction was observed to have a reasonably good prediction accuracy compared to the standard RNN prediction model. This was discovered after it was suggested. Yet, the only things that were taken into consideration by the models' input variables were the temperature of the air outside and the number of hours in a day. They indicated that more input variables for overcast day circumstances, including cloud coverage, sunlight duration, and other similar factors, would be considered in the future.

5. Challenges for ML in materials sciences

5.1. Data acquisition

ML needs big data for training, and it must be pervasive and high-quality. There may also be times when it is appropriate to wait for new data production. A large volume of data is needed to achieve high accuracy. There are millions of data points in some areas, such as image processing. However, input parameters are limited in the field of energy materials, and the number of input data points is usually 100–1000. The larger the volume of data, the higher the accuracy.

5.2. Time and resources

ML requires enough time so that the algorithms can learn and develop their goals carefully. Therefore, ML requires considerable resources to operate. One of the challenges ML faces is comparing the anticipated results with experimental data, which requires considerable time and energy.

5.3. Interpretation of results and more software expansion

Another critical challenge is the accuracy of the results produced by algorithms. Algorithms should also be carefully selected according to their goals. Working with ML software now requires strong programming ability and deep experience in the field. Errors are very likely a strong computer programming background is required. Most people who work in the energy materials field are experimental and need more knowledge of computer programming. Therefore, more straightforward, easily used applications are needed that allow users to obtain accurate results.

5.4. Cost and high sensitivity of error

Usually, ML projects require software infrastructure, which can be costly. ML can be highly susceptible to error; for example, if the data input to an algorithm for a small solar cell is not pervasive, we will be confronted with false forecasts, and this error will continue going forward. When the error is noticed, the source may not be known, and may take a long time to correct.

6. Conclusions

Today, ML applications in energy materials science are the most reliable method for predicting materials and system behavior.

Selecting a suitable algorithm for using ML is always challenging, and examining the different algorithms requires much time. In this review paper, we investigated and compared the most recent published papers, and according to the obtained data, the most used algorithm was ANN (MLP) for solar cells, batteries and PCMs. This is probably because the number of input data points in energy materials is usually limited to between a hundred and a thousand, and the MLP algorithm is suitable for networks with limited data. Base neural networks such as MLP have less separation power than robust networks such as SVM. However, when learning an MLP network by adding hidden layers, using a strong activation function and choosing the correct number of hidden layers neurons in deep learning, such a network can even better do better than a backup vector machine. An MLP network with

numerous hidden layers can maintain its power in high class-number situations such as five classes if an ordinary network (with fewer hidden layers) MLP loses power to some extent when there are more than two classes.

ML has exhibited a lot of relevance in processing and storage of informative data on the various characteristics of different energy materials. Though some gaps still exist, researchers are persistently searching for new and viable information using affordable ML techniques.

We have thoroughly examined the ML algorithms used in energy material science in recent years and obtained each algorithm's error rate and accuracy. Since ML has been widely used in materials science recently and choosing a suitable algorithm for predicting material energy behavior is vital, we investigated reports of perovskite solar cells, batteries, and PCMs and listed the algorithms used and the error rate of each algorithm. In general, since the volume of input data of energy materials is consistently low (100–100,000 data points), in order to have high accuracy in the results, ANN is the best algorithm, and SVM is also widely used in energy materials science.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] D.T. Tran, et al., Heterogeneous multilayer generalized operational perceptron, IEEE Trans. Neural Netw. Learn. Syst. 31 (3) (2019) 710–724.
[2] N. Brown, T. Sandholm, Superhuman AI for multiplayer poker, Science 365 (6456) (2019) 885–890.
[3] J. Pei, et al., Towards artificial general intelligence with hybrid Tianjic chip architecture, Nature 572 (7767) (2019) 106–111.
[4] Y. Yao, et al., Sensing spatial distribution of urban land use by integrating points-of-interest and Google Word2Vec model, Int. J. Geogr. Inf. Sci. 31 (4) (2017) 825–848.
[5] A.W. Senior, et al., Improved protein structure prediction using potentials from deep learning, Nature 577 (7792) (2020) 706–710.
[6] M. Popova, O. Isayev, A. Tropsha, Deep reinforcement learning for de novo drug design, Sci. Adv. 4 (7) (2018) eaap7885.
[7] D. Zhang, R. Cao, S. Wu, Information fusion in visual question answering: a survey, Inf. Fusion 52 (2019) 268–280.
[8] Y. Liu, et al., Inkjet-printed unclonable quantum dot fluorescent anti-counterfeiting labels with artificial intelligence authentication, Nat. Commun. 10 (1) (2019) 1–9.
[9] Y. LeCun, Y. Bengio, G. Hinton, Deep learning, nature 521 (7553) (2015) 436–444.
[10] I.A.T. Hashem, et al., The role of big data in smart city, Int. J. Inf. Manag. 36 (5) (2016) 748–758.
[11] M.S. Yadav, P.A. Pavlou, Marketing in computer-mediated environments: Research synthesis and new directions, J. Mark. (2014).
[12] C. Nantasenamat, C. Isarankura-Na-Ayudhya, V. Prachayasittikul, Advances in computational methods to predict the biological activity of compounds, Expert Opin. Drug Discov. 5 (7) (2010) 633–654.
[13] S. Sivasankaran, et al., Analysis of workability behavior of Al–SiC P/M composites using backpropagation neural network model and statistical technique, Comput. Mater. Sci. 47 (1) (2009) 46–59.
[14] B. Schölkopf, K. Tsuda, J.-P. Vert, Kernel Methods in Computational Biology, MIT press, 2004.
[15] L. Devroye, L. Györfi, G. Lugosi, A Probabilistic Theory of Pattern Recognition Vol. 31 Springer Science & Business Media, 2013.
[16] Boser, B.E., I.M. Guyon, and V.N. Vapnik. A training algorithm for optimal margin classifiers. in Proceedings of the fifth annual workshop on Computational learning theory. 1992.
[17] J. Fan, et al., Empirical and machine learning models for predicting daily global solar radiation from sunshine duration: A review and case study in China, Renew. Sustain. Energy Rev. 100 (2019) 186–212.
[18] Y. Zhou, S. Zheng, G. Zhang, Machine learning-based optimal design of a phase change material integrated renewable system with on-site PV, radiative cooling and hybrid ventilations—study of modelling and application in five climatic regions, Energy 192 (C) (2020) p. S0360544219323035.
[19] J. Kirman, et al., Machine-learning-accelerated perovskite crystallization, Matter 2 (4) (2020) 938–947.
[20] N.T.P. Hartono, et al., How machine learning can help select capping layers to suppress perovskite degradation, Nat. Commun. 11 (1) (2020) 4172.
[21] S. Zhang, et al., Predicting the formability of hybrid organic–inorganic perovskites via an interpretable machine learning strategy, J. Phys. Chem. Lett. 12 (31) (2021) 7423–7430.
[22] M. Srivastava, et al., Machine learning roadmap for perovskite photovoltaics, J. Phys. Chem. Lett. 12 (32) (2021) 7866–7877.
[23] Y. Yu, et al., Machine learning for understanding compatibility of organic–inorganic hybrid perovskites with post-treatment amines, ACS Energy Lett. 4 (2) (2019) 397–404.
[24] S. Lu, et al., Accelerated discovery of stable lead-free hybrid organic-inorganic perovskites via machine learning, Nat. Commun. 9 (1) (2018) 3405.
[25] Y. Liu, et al., Machine learning assisted materials design and discovery for rechargeable batteries, Energy Storage Mater. 31 (2020) 434–450.
[26] N.J. Kelly, P.G. Tuohy, A.D. Hawkes, Performance assessment of tariff-based air source heat pump load shifting in a UK detached dwelling featuring phase change-enhanced buffering, Appl. Therm. Eng. 71 (2) (2014) 809–820.
[27] Y. Zhou, S. Zheng, G. Zhang, A review on cooling performance enhancement for phase change materials integrated systems—flexible design and smart control with machine learning applications, Build. Environ. 174 (2020) 106786.
[28] A. Mahmood, J.-L. Wang, Machine learning for high performance organic solar cells: current scenario and future prospects, Energy Environ. Sci. 14 (1) (2021) 90–105.
[29] Y. Liu, et al., Materials discovery and design using machine learning, J. Mater. 3 (3) (2017) 159–177.
[30] W.J. Murdoch, et al., Definitions, methods, and applications in interpretable machine learning, Proc. Natl. Acad. Sci. 116 (44) (2019) 22071–22080.
[31] S.B. Kotsiantis, I. Zaharakis, P. Pintelas, Supervised machine learning: a review of classification techniques, Emerg. Artif. Intell. Appl. Comput. Eng. 160 (1) (2007) 3–24.
[32] Mitchell, T.M., The discipline of machine learning. Vol. 9. 2006: Carnegie Mellon University, School of Computer Science, Machine Learning ….
[33] A. L'heureux, et al., Machine learning with big data: challenges and approaches, IEEE Access 5 (2017) 7776–7797.
[34] C. Farabet, et al., Learning hierarchical features for scene labeling, IEEE Trans. Pattern Anal. Mach. Intell. 35 (8) (2012) 1915–1929.
[35] M.J. Azur, et al., Multiple imputation by chained equations: what is it and how does it work? Int. J. Methods Psychiatr. Res. 20 (1) (2011) 40–49.
[36] J.V. Tu, Advantages and disadvantages of using artificial neural networks versus logistic regression for predicting medical outcomes, J. Clin. Epidemiol. 49 (11) (1996) 1225–1231.
[37] P. Rzepakowski, S. Jaroszewicz, Decision trees for uplift modeling with single and multiple treatments, Knowl. Inf. Syst. 32 (2) (2012) 303–327.
[38] A. Géron, Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media, Inc, 2022.
[39] Hoermann, S., M. Bach, and K. Dietmayer. Dynamic occupancy grid prediction for urban autonomous driving: A deep learning approach with fully automatic labeling. in 2018 IEEE International Conference on Robotics and Automation (ICRA). 2018. IEEE.
[40] Ide, N. and J. Véronis. Extracting knowledge bases from machine-readable dictionaries: Have we wasted our time. in KB&KS Workshop. 1993.
[41] J. Dean, et al., Large scale distributed deep networks, Adv. Neural Inf. Process. Syst. (2012) **25**.
[42] J. Schmidt, et al., Recent advances and applications of machine learning in solid-state materials science, npj Comput. Materials 5 (1) (2019) 1–36.
[43] Nataraj, L., et al. A comparative assessment of malware classification using binary texture analysis and dynamic analysis. in Proceedings of the 4th ACM Workshop on Security and Artificial Intelligence. 2011.
[44] M.C. Hout, M.H. Papesh, S.D. Goldinger, Multidimensional scaling, Wiley Interdiscip. Rev.: Cogn. Sci. 4 (1) (2013) 93–103.
[45] M. Usama, et al., Unsupervised machine learning for networking: Techniques, applications and research challenges, IEEE Access 7 (2019) 65579–65615.
[46] Doulamis, N. and A. Voulodimos. FAST-MDL: Fast Adaptive Supervised Training of multi-layered deep learning models for consistent object tracking and classification. in 2016 IEEE International Conference on Imaging Systems and Techniques (IST). 2016. IEEE.

[47] Bhowmik, P., Machine Learning in Production: From Experimented ML Model to System. J Robot Auto Res, 3 (2), 200–208.
[48] Pritom Bhowmik B. Tech.(Computer Science & Engineering) In-stitute of Engineering & Management, Kolkata, India. B. Tech.(Computer Science & Engineering) Institute of Engineering & Management, Kolkata, India ISSN, 2022. 2831: p. 6789.
[49] S. Schelter, J. Stoyanovich, Taming technical bias in machine learning pipelines, Bull. Tech. Comm. Data Eng. 43 (4) (2020).
[50] Ziebart, B.D., et al. Maximum entropy inverse reinforcement learning. in Aaai. 2008. Chicago, IL, USA.
[51] S.D. Whitehead, D.H. Ballard, Learning to perceive and act by trial and error, Mach. Learn. 7 (1) (1991) 45–83.
[52] P. Langley, Elements of Machine Learning, Morgan Kaufmann, 1996.
[53] R.S. Sutton, A.G. Barto, Reinforcement Learning: An Introduction, MIT press, 2018.
[54] T. Back, Evolutionary algorithms in theory and practice: evolution strategies, evolutionary programming, genetic algorithms, Oxford university press, 1996.
[55] M. Hutter, Universal artificial intelligence: Sequential decisions based on algorithmic probability, Springer Science & Business Media, 2004.
[56] A.J. Stimpson, A machine learning approach to modeling and predicting training effectiveness, Massachusetts Institute of Technology, 2015.
[57] Bontrager, P.J., Learning to Generate Form and Function. 2020, New York University Tandon School of Engineering.
[58] Z. Yang, et al., An efficient destination prediction approach based on future trajectory prediction and transition matrix optimization, IEEE Trans. Knowl. data Eng. 32 (2) (2018) 203–217.
[59] Hoxmark, B. and J. Wilhelmsen, *Domain general Active Learning strategies using inter-sample similarity and Reinforcement Learning*. 2018, NTNU.
[60] R. Feldman, J. Sanger, The text mining handbook: advanced approaches in analyzing unstructured data, Cambridge university press, 2007.
[61] R.K. Ando, T. Zhang, P. Bartlett, A framework for learning predictive structures from multiple tasks and unlabeled data, J. Mach. Learn. Res. 6 (11) (2005).
[62] C.-F. Chien, Y.-S. Lin, S.-K. Lin, Deep reinforcement learning for selecting demand forecast models to empower Industry 3.5 and an empirical study for a semiconductor component distributor, Int. J. Prod. Res. 58 (9) (2020) 2784–2804.
[63] D.V. Carvalho, E.M. Pereira, J.S. Cardoso, Machine learning interpretability: A survey on methods and metrics, Electronics 8 (8) (2019) 832.
[64] R.S. Sutton, A.G. Barto, Reinforcement learning, J. Cogn. Neurosci. 11 (1) (1999) 126–134.
[65] P. Brazdil, et al., Metalearning: Applications to data mining, Springer Science & Business Media, 2008.
[66] L. Yang, A. Shami, On hyperparameter optimization of machine learning algorithms: Theory and practice, Neurocomputing 415 (2020) 295–316.
[67] A. Seko, A. Togo, I. Tanaka, Descriptors for machine learning of materials, in: I. Tanaka (Ed.), *Data*, in *Nanoinformatics*, Springer Singapore, Singapore, 2018, pp. 3–23.
[68] M. Amini, K. Sharifani, A. Rahmani, Machine learning model towards evaluating data gathering methods in manufacturing and mechanical engineering, Int. J. Appl. Sci. Eng. Res. 2023 (15) (2023) 349–362.
[69] P. Patel, S.P. Ong, Artificial intelligence is aiding the search for energy materials, MRS Bull. 44 (3) (2019) 162–163.
[70] P. Ball, Using artificial intelligence to accelerate materials development, Mrs Bull. 44 (5) (2019) 335–344.
[71] M.E. Celebi, K. Aydin, Unsupervised learning algorithms Vol. 9 Springer, 2016.
[72] Khatib, M.E. and W.A. de Jong, ML4Chem: A Machine Learning Package for Chemistry and Materials Science. arXiv preprint arXiv:2003.13388, 2020.
[73] C.P. Gomes, B. Selman, J.M. Gregoire, Artificial intelligence for materials discovery, MRS Bull. 44 (7) (2019) 538–544.
[74] H. Liu, et al., Dynamic nonlinear partial least squares modeling using gaussian process regression, Ind. Eng. Chem. Res. 58 (36) (2019) 16676–16686.
[75] Wang, J., An intuitive tutorial to Gaussian processes regression. arXiv preprint arXiv:2009.10862, 2020.
[76] A. Chen, X. Zhang, Z. Zhou, Machine learning: accelerating materials development for energy storage and conversion, InfoMat 2 (3) (2020) 553–576.
[77] A. Schneider, G. Hommel, M. Blettner, Linear regression analysis: part 14 of a series on evaluation of scientific publications, Dtsch. Ärzteblatt Int. 107 (44) (2010) 776.
[78] A.P. Dempster, M. Schatzoff, N. Wermuth, A simulation study of alternatives to ordinary least squares, J. Am. Stat. Assoc. 72 (357) (1977) 77–91.
[79] L. Himanen, et al., Data-driven materials science: status, challenges, and perspectives. Adv. Sci. 6 (21) (2019) 1900808.
[80] M. Maalouf, Logistic regression in data analysis: an overview, Int. J. Data Anal. Tech. Strateg. 3 (3) (2011) 281–299.
[81] B. Wang, et al., Analysis of passenger's choice between shuttle bus and illegal taxi, Procedia-Soc. Behav. Sci. 96 (2013) 1948–1960.
[82] A. Upadhyay, G. Bandyopadhyay, A. Dutta, Forecasting stock performance in indian market using multinomial logistic regression, J. Bus. Stud. Q. 3 (3) (2012) 16.
[83] A.D. Sendek, et al., Holistic computational structure screening of more than 12 000 candidates for solid lithium-ion conductor materials, Energy Environ. Sci. 10 (1) (2017) 306–320.
[84] L. Zhao, Y. Chen, D.W. Schaffner, Comparison of logistic regression and linear regression in modeling percentage data, Appl. Environ. Microbiol. 67 (5) (2001) 2129–2135.
[85] J. Lever, M. Krzywinski, N. Altman, Logistic regression, Nat. Methods 13 (7) (2016) 541–542.
[86] M. Hellström, J. Behler, Neural network potentials in materials modeling, Handb. Mater. Model.: Methods.: Theory Model. (2020) 661–680.
[87] H. Bhadeshia, et al., Performance of neural networks in materials science, Mater. Sci. Technol. 25 (4) (2009) 504–510.
[88] W. Sha, K. Edwards, The use of artificial neural networks in materials science based research, Mater. Des. 28 (6) (2007) 1747–1752.
[89] Y. Sun, G.G. Yen, Z. Yi, Evolving unsupervised deep neural networks for learning meaningful representations, IEEE Trans. Evolut. Comput. 23 (1) (2018) 89–103.
[90] K. Gopalakrishnan, et al., Deep convolutional neural networks with transfer learning for computer vision-based data-driven pavement distress detection, Constr. Build. Mater. 157 (2017) 322–330.
[91] Heidaryan, M. and F. Farokhi. Robust hand gestures tracking method in cluttered background based on multilayer perceptron. in 2015 5th International Conference on Computer and Knowledge Engineering (ICCKE). 2015. IEEE.
[92] B.M.S. Hasan, A.M. Abdulazeez, A review of principal component analysis algorithm for dimensionality reduction, J. Soft Comput. Data Min. 2 (1) (2021) 20–30.
[93] Al-Aidaroos, K.M., A.A. Bakar, Z. Othman. Naive Bayes variants in classification learning. in 2010 international conference on inform*ation retrieval & knowledge management (CAMP)*. 2010. IEEE.
[94] Y. Zhang, et al., The prediction of spark-ignition engine performance and emissions based on the SVR algorithm, Processes 10 (2022), https://doi.org/10.3390/pr10020312
[95] P.O. Gislason, J.A. Benediktsson, J.R. Sveinsson, Random forests for land cover classification, Pattern Recognit. Lett. 27 (4) (2006) 294–300.
[96] A.J. Myles, et al., An introduction to decision tree modeling, J. Chemom.: A J. Chemom. Soc. 18 (6) (2004) 275–285.
[97] Van Hasselt, H., A. Guez, and D. Silver. Deep reinforcement learning with double q-learning. in Proceedings of the AAAI conference on artificial intelligence. 2016.
[98] K.T. Butler, et al., Machine learning for molecular and materials science, Nature 559 (7715) (2018) 547–555.
[99] J.P. Holdren, P.R. Ehrlich, Human Population and the Global Environment: Population growth, rising per capita material consumption, and disruptive technologies have made civilization a global ecological force, Am. Sci. 62 (3) (1974) 282–292.
[100] N. Afsharzade, et al., Renewable energy development in rural areas of Iran, Renew. Sustain. Energy Rev. 65 (2016) 743–755.
[101] G.K. Singh, Solar power generation by PV (photovoltaic) technology: a review, Energy 53 (2013) 1–13.
[102] Z. Liu, et al., Machine learning with knowledge constraints for process optimization of open-air perovskite solar cell manufacturing, Joule 6 (4) (2022) 834–849.
[103] L. Cheng, T. Yu, A new generation of AI: a review and perspective on machine learning technologies applied to smart energy and electric power systems, Int. J. Energy Res. 43 (6) (2019) 1928–1973.
[104] M.H. Lee, Insights from machine learning techniques for predicting the efficiency of fullerene derivatives-based ternary organic solar cells at ternary blend design, Adv. Energy Mater. 9 (26) (2019) 1900891.
[105] E.V. Kerschaver, G. Beaucarne, Back-contact solar cells: a review. Prog. Photovolt.: Res. Appl. 14 (2) (2006) 107–123.
[106] A.J. Nozik, et al., Semiconductor quantum dots and quantum dot arrays and applications of multiple exciton generation to third-generation photovoltaic solar cells, Chem. Rev. 110 (11) (2010) 6873–6890.
[107] D. Yang, D. Ma, Development of organic semiconductor photodetectors: from mechanism to applications, Adv. Opt. Mater. 7 (1) (2019) 1800522.
[108] J. Li, et al., Machine-learning-assisted discovery of perovskite materials with high dielectric breakdown strength, Mater. Adv. 3 (23) (2022) 8639–8646.
[109] T. Banerjee, et al., Polymer photocatalysts for solar-to-chemical energy conversion, Nature Reviews Materials 6 (2) (2021) 168–190.
[110] Y. Hu, et al., Machine-learning modeling for ultra-stable high-efficiency perovskite solar cells, Adv. Energy Mater. 12 (41) (2022) 2201463.
[111] X. Zhai, M. Chen, W. Lu, Accelerated search for perovskite materials with higher Curie temperature based on the machine learning methods, Comput. Mater. Sci. 151 (2018) 41–48.
[112] X. Cai, et al., Data-driven design of high-performance MASnxPb1-xI3 perovskite materials by machine learning and experimental realization, Light.: Sci. Appl. 11 (1) (2022) 234.
[113] Z. Guo, B. Lin, Machine learning stability and band gap of lead-free halide double perovskite materials for perovskite solar cells, Sol. Energy 228 (2021) 689–699.
[114] Q. Tao, et al., Machine learning aided design of perovskite oxide materials for photocatalytic water splitting, J. Energy Chem. 60 (2021) 351–359.
[115] Q. Xu, et al., Rationalizing perovskite data for machine learning and materials design, J. Phys. Chem. Lett. 9 (24) (2018) 6948–6954.
[116] H.-A. Chen, et al., Microstructure maps of complex perovskite materials from extensive Monte Carlo sampling using machine learning enabled energy model, J. Phys. Chem. Lett. 12 (14) (2021) 3591–3599.
[117] Z. Pan, Y. Zhou, L. Zhang, Photoelectrochemical properties, machine learning, and symbolic regression for molecularly engineered halide perovskite materials in water, ACS Appl. Mater. Interfaces 14 (7) (2022) 9933–9943.
[118] S. Jarin, et al., Predicting the crystal structure and lattice parameters of the perovskite materials via different machine learning models based on basic atom properties, Crystals 12 (11) (2022) 1570.
[119] C. Li, et al., Improved physics-based structural descriptors of perovskite materials enable higher accuracy of machine learning, Comput. Mater. Sci. 198 (2021) 110714.
[120] N. Filipoiu, et al., Investigation of opto-electronic properties and stability of mixed-cation mixed-halide perovskite materials with machine-learning implementation, Energies (2021) **14**, https://doi.org/10.3390/en14175431
[121] G. Crabtree, The coming electric vehicle transformation, Science 366 (6464) (2019) 422–424.

[122] M. Isik, R. Dodder, P.O. Kaplan, Transportation emissions scenarios for New York City under different carbon intensities of electricity and electric vehicle adoption rates, Nat. Energy 6 (1) (2021) 92–104.
[123] M. Berecibar, Accurate predictions of lithium-ion battery life, Nature 568 (7752) (2019) 325–326.
[124] X. Han, et al., A review on the key issues of the lithium ion battery degradation among the whole life cycle, ETransportation 1 (2019) 100005.
[125] Y. Gao, et al., Aging mechanisms under different state-of-charge ranges and the multi-indicators system of state-of-health for lithium-ion battery with Li (NiMnCo) O2 cathode, J. Power Sources 400 (2018) 641–651.
[126] L. Wildfeuer, M. Lienkamp, Quantifiability of inherent cell-to-cell variations of commercial lithium-ion batteries, ETransportation 9 (2021) 100129.
[127] Y. Lu, et al., A method of cell-to-cell variation evaluation for battery packs in electric vehicles with charging cloud data, ETransportation 6 (2020) 100077.
[128] W. Chen, et al., A review of lithium-ion battery for electric vehicle applications and beyond, Energy Procedia 158 (2019) 4363–4368.
[129] Z. Yang, et al., Machine learning accelerates the discovery of light-absorbing materials for double perovskite solar cells, J. Phys. Chem. C. 125 (41) (2021) 22483–22492.
[130] D. Wang, et al., *I*nterfaces in garnet-based all-solid-state lithium batteries. Adv. Energy Materials 10 (39) (2020) 2001318.
[131] H. Lee, et al., Advances and prospects of sulfide all-solid-state lithium batteries via one-to-one comparison with conventional liquid lithium ion batteries, Adv. Mater. 31 (29) (2019) 1900376.
[132] N. Boaretto, et al., Lithium solid-state batteries: State-of-the-art and challenges for materials, interfaces and processing, J. Power Sources 502 (2021) 229919.
[133] W.F. Maier, K. Stoewe, S. Sieg, Combinatorial and high-throughput materials science. Angew. Chem. Int. Ed. 46 (32) (2007) 6016–6067.
[134] A. Jain, et al., Commentary: the materials project: a materials genome approach to accelerating materials innovation, APL Mater. 1 (1) (2013) 011002.
[135] B. Esser, et al., A perspective on organic electrode materials and technologies for next generation batteries, J. Power Sources 482 (2021) 228814.
[136] Y. Tian, et al., Promises and challenges of next-generation "beyond Li-ion" batteries for electric vehicles and grid decarbonization, Chem. Rev. 121 (3) (2020) 1623–1669.
[137] C. Lv, et al., Machine learning: an advanced platform for materials development and state prediction in lithium-ion batteries. Adv. Mater. 34 (25) (2022) 2101474.
[138] S.S. Ng, Y. Xing, K.L. Tsui, A naive Bayes model for robust remaining useful life prediction of lithium-ion battery, Appl. Energy 118 (2014) 114–123.
[139] Liu, Z., et al. Prediction of remaining useful life of battery cell using logistic regression based on strong tracking particle filter. in 2015 IEEE Conference on Prognostics and Health Management (PHM). 2015. IEEE.
[140] H.-M. Cheng, et al., Developing multivariate linear regression models to predict the electrochemical performance of lithium ion batteries based on material property parameters, J. Electrochem. Soc. 164 (7) (2017) A1393.
[141] Y. Zhou, M. Huang, M. Pecht, Remaining useful life estimation of lithium-ion cells based on k-nearest neighbor regression with differential evolution optimization, J. Clean. Prod. 249 (2020) 119409.
[142] Z. Zhang, et al., State-of-health estimation for the lithium-ion battery based on gradient boosting decision tree with autonomous selection of excellent features. Int. J. Energy Res. 46 (2) (2022) 1756–1765.
[143] B. Wu, et al., Application of artificial neural networks in design of lithium-ion batteries, J. Power Sources 395 (2018) 128–136.
[144] Cailian, L. Life prediction of battery based on random forest optimized by genetic algorithm. in 2020 IEEE International Conference on Prognostics and Health Management (ICPHM). 2020. IEEE.
[145] S. Saxena, et al., A convolutional neural network model for battery capacity fade curve prediction using early life data, J. Power Sources 542 (2022) 231736.
[146] J. Lu, et al., Battery degradation prediction against uncertain future conditions with recurrent neural network enabled deep learning, Energy Storage Mater. (2022).
[147] Y. Zhou, W. Chuck, G. Zhang, Study on heat-transfer mechanism of wallboards containing active phase change material and parameter optimization with ventilation, Appl. Therm. Eng. 144 (2018) 1091–1108.
[148] Y. Zhou, et al., Thermal performance and optimized thickness of active shape-stabilized PCM boards for side-wall cooling and under-floor heating system, Indoor Built Environ. 25 (8) (2016) 1279–1295.
[149] T. Yan, et al., Simulation study of a pipe-encapsulated PCM wall system with self-activated heat removal by nocturnal sky radiation, Renew. Energy 146 (2020) 1451–1464.
[150] F. Ma, P. Zhang, Performance investigation of the direct absorption solar collector based on phase change slurry, Appl. Therm. Eng. 162 (2019) 114244.
[151] Y. Zhou, S. Zheng, G. Zhang, Multivariable optimisation of a new PCMs integrated hybrid renewable system with active cooling and hybrid ventilations, J. Build. Eng. 26 (2019) 100845.
[152] Y. Zhou, S. Zheng, G. Zhang, Artificial neural network based multivariable optimization of a hybrid system integrated with phase change materials, active cooling and hybrid ventilations, Energy Convers. Manag. 197 (2019) 111859.
[153] K. Kant, et al., Analysis and design of air ventilated building integrated photovoltaic (BIPV) system incorporating phase change materials, Energy Convers. Manag. 196 (2019) 149–164.
[154] J.C.A. Anton, et al., Support vector machines used to estimate the battery state of charge, IEEE Trans. Power Electron. 28 (12) (2013) 5919–5926.
[155] R. Kandasamy, X.Q. Wang, A.S. Mujumdar, Transient cooling of electronics using phase change material (PCM)-based heat sinks, Appl. Therm. Eng. 28 (8–9) (2008) 1047–1057.
[156] J. Zhao, C. Wu, Z. Rao, Investigation on the cooling and temperature uniformity of power battery pack based on gradient phase change materials embedded thin heat sinks, Appl. Therm. Eng. 174 (2020).
[157] A. Sharma, et al., Review on thermal energy storage with phase change materials and applications, Renew. Sustain. Energy Rev. 13 (2) (2009) 318–345.
[158] G.V.S. Anooj, G.K. Marri, C. Balaji, A machine learning methodology for the diagnosis of phase change material-based thermal management systems, Appl. Therm. Eng. 222 (2023) 119864.
[159] S. Abbasian-Naghneh, R. Kalbasi, Implementation of ANN and GA on building with PCM at various setpoints, PCM types, and installation locations to boost energy saving and CO_2 saving, Eng. Anal. Bound. Elem. 144 (2022) 110–126.
[160] L. Tang, et al., Exergy-based optimisation of a phase change materials integrated hybrid renewable system for active cooling applications using supervised machine learning method, Sol. Energy 195 (2020) 514–526.
[161] Y. Gao, et al., Thermophysical properties prediction of carbon-based nano-enhanced phase change material's using various machine learning methods, J. Taiwan Inst. Chem. Eng. (2023) 104662.
[162] G. Ren, A. Chuttar, D. Banerjee, Exploring efficacy of machine learning (artificial neural networks) for enhancing reliability of thermal energy storage platforms utilizing phase change materials, Int. J. Heat. Mass Transf. 189 (2022) 122628.
[163] A.T. Fini, A. Fattahi, S. Musavi, Machine learning prediction and multiobjective optimization for cooling enhancement of a plate battery using the chaotic water-microencapsulated PCM fluid flows, J. Taiwan Inst. Chem. Eng. (2023) 104680.
[164] Ebrahimpour, M., et al. Optimal control for An Active Phase Change Material System Using Reinforcement Learning. in 2022 IEEE International Symposium on Advanced Control of Industrial Processes (AdCONIP). 2022.
[165] A. Bamasag, et al., Machine learning-based prediction and augmentation of dish solar distiller performance using an innovative convex stepped absorber and phase change material with nanoadditives, Process Saf. Environ. Prot. 162 (2022) 112–123.
[166] M.O. Karaağaç, et al., Experimental analysis of CPV/T solar dryer with nano-enhanced PCM and prediction of drying parameters using ANN and SVM algorithms, Sol. Energy 218 (2021) 57–67.
[167] T. Tietze, et al., Application of phase change material and artificial neural networks for smoothing of heat flux fluctuations, Energies 14 (12) (2021) 3531.
[168] S. Ibrahim, et al., Linear regression model in estimating solar radiation in perlis, Energy Procedia 18 (2012) 1402–1412.
[169] V. Jagadeesh, K. Venkata Subbaiah, J. Varanasi, Forecasting the probability of solar power output using logistic regression algorithm, J. Stat. Manag. Syst. 23 (1) (2020) 1–16.
[170] Azza, M., et al., *Automatic solution for solar cell photo-current prediction using machine learning*. E3S Web Conf., 2021. **297**: p. 01029.
[171] Z. Pang, F. Niu, Z. O'Neill, Solar radiation prediction using recurrent neural network and artificial neural network: a case study with comparisons, Renew. Energy 156 (2020) 279–289.
[172] J.C. Álvarez Antón, et al., Battery state-of-charge estimator using the SVM technique, Appl. Math. Model. 37 (9) (2013) 6244–6253.
[173] Harippriya, S., E.E. Vigneswaran, and S. Jayanthy. Battery Management System to Estimate Battery Aging using Deep Learning and Machine Learning Algorithms. in Journal of Physics: Conference Series. 2022. IOP Publishing.
[174] M.-H. Lee, Robust random forest based non-fullerene organic solar cells efficiency prediction, Org. Electron. 76 (2020) 105465.
[175] S. Behara, T. Poonawala, T. Thomas, Crystal structure classification in ABO3 perovskites via machine learning, Comput. Mater. Sci. 188 (2021) 110191.
[176] D.C. Nguyen, Y. Ishikawa, Artificial neural network for predicting annual output energy of building-integrated photovoltaics based on the 2-terminal perovskite/silicon tandem cells under realistic conditions, Energy Rep. 8 (2022) 10819–10832.
[177] Y. Onal, Gaussian kernel based SVR model for short-term photovoltaic MPP power prediction, Comput. Syst. Sci. Eng. 41 (1) (2022) 141–156.
[178] R. Ranjan, B. Huang, A. Fatehi, Robust Gaussian process modeling using EM algorithm, J. Process Control 42 (2016) 125–136.
[179] A. Ihom, A. Offiong, Neural networks in materials science and engineering: a review of salient issues, Eur. J. Eng. Technol. 3 (7) (2015) 40–54.
[180] K. Suzuki, et al., Fast material search of lithium ion conducting oxides using a recommender system, J. Mater. Chem. A 8 (23) (2020) 11582–11588.
[181] W.-C. Lu, et al., Using support vector machine for materials design, Adv. Manuf. 1 (2013) 151–159.
[182] K. Sodeyama, et al., Liquid electrolyte informatics using an exhaustive search with linear regression, Phys. Chem. Chem. Phys. 20 (35) (2018) 22585–22591.
[183] Igarashi, Y., et al. *Three levels of data-driven science*. in Journal of Physics: Conference Series. 2016. IOP Publishing.
[184] Wang, Y., et al. Comparisons and selections of features and classifiers for short text classification. in Iop conference series: Materials science and engineering. 2017. IOP Publishing.
[185] Mitrofanov, S. and E. Semenkin. Tree retraining in the decision tree learning algorithm. in IOP Conference Series: Materials Science and Engineering. 2021. IOP Publishing.